

AI-Driven Analysis of the Kidneys from CT Scans

Bachelor Project

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Abstract

The kidneys are vital organs responsible for blood filtration. Chronic kidney disease (CKD) is diagnosed when kidney damage and/or a severely reduced glomerular filtration rate (GFR) persist for at least three months and is a serious, widespread condition associated with increased cardiovascular risk and high mortality. Because CKD and donor suitability are linked to structural changes in the kidneys and their substructures, accurate segmentation of renal anatomy is essential for clinical decision making and research. However, manual segmentation is time-consuming, motivating the development of automated methods.

In this project, an AI-driven framework for cortex–medulla–cyst segmentation was implemented using state-of-the-art deep learning models. The framework was evaluated quantitatively on a small dataset with manually annotated ground truth and visually on a larger dataset including healthy subjects and patients with low eGFR or hypertension. Volumes were extracted from the obtained segmentations, and statistical tests assessed differences between low eGFR patients and controls. Additionally, contrast phases were predicted using the TotalSegmentator tool, and an nnU-Net for cortex-medulla-cyst segmentation was trained on pseudo labels.

The framework achieved high Dice similarity coefficients (DSCs) on the small dataset, consistently higher for cortex than medulla (mean 0.90 vs 0.82, excluding an outlier). Segmentations on the larger dataset were generally realistic, though some imperfections were observed, likely due to differences in contrast phase, age, kidney function, and structural abnormalities. Predicted contrast phases aligned well with protocol expectations, and the late arterial phase seemed more variable and slightly harder to predict. Statistically significant differences were found between low eGFR patients and controls for cortex, medulla, and cyst volumes. The nnU-Net trained on pseudo ground truth achieved stable DSCs across folds for cortex and medulla (mean 0.93 and 0.90), while cyst segmentation remained challenging due to class imbalance (mean DSC 0.2).

These results demonstrate the potential of AI-based kidney substructure segmentation to enable large-scale volumetric analysis, while highlighting the need for larger, more balanced datasets with high-quality ground truth to improve segmentation performance, robustness, and the ability to attribute observed volumetric differences between groups specifically to eGFR.

Contents

1	Introduction	1
2	Clinical Background	2
2.1	Renal Anatomy and Physiology	2
2.2	Diseases Affecting the Kidneys	5
2.3	Computed Tomography	6
3	Technical Background and Methods	9
3.1	Neural Networks	9
3.1.1	Convolutional Neural Networks	10
3.1.2	Training a Neural Network	11
3.2	Gradient Tree Boosting	12
3.2.1	Regression trees	13
3.2.2	Tree Boosting	13
3.2.3	Gradient Tree Boosting	13
3.3	Evaluation Metrics	16
3.4	Statistical Hypothesis Testing	16
4	State-of-the-art	19
4.1	Kidney Segmentation from CT Scans	19
4.1.1	Tang et al. (2021)	20
4.1.2	Korfiatis et al. (2022)	20
4.1.3	Yu et al. (2023)	21
4.1.4	Chen et al. (2025)	22
5	Dataset Description	24
5.1	Data provided by TotalSegmentator	24
5.2	Data provided by Rigshospitalet	25
6	Experiments and Results	27
6.1	Cortex–Medulla–Cyst Segmentation	27
6.1.1	Models	27
6.1.2	Framework	28
6.1.3	Results	29
6.2	Contrast Phase Detection	32

6.2.1	Model	32
6.2.2	Results	33
6.3	Volumetric Biomarker Analysis	34
6.4	Segmentation Using nnU-Net	37
6.4.1	Model	37
6.4.2	Results	38
7	Discussion	39
7.1	Future Work	42
8	Conclusion	44
	References	46
A	Appendix	51
A.1	UNesT Model Architecture	51
A.2	Distribution of Post-Injection Times	52

Abbreviations

Abbreviation	Description
GFR	Glomerular Filtration Rate
eGFR	Estimated Glomerular Filtration Rate
PKD	Polycystic Kidney Disease
CKD	Chronic Kidney Disease
ESKD	End-Stage Kidney Disease
AKI	Acute Kidney Injury
CT	Computed Tomography
HU	Hounsfield Unit
NN	Neural Network
MLP	Multi-Layer Perceptron
CNN	Convolutional Neural Network
ReLU	Rectified Linear Unit
CE	Cross-Entropy
DSC	Dice Score
ViT	Vision Transformer
ROI	Region of Interest

CHAPTER 1

Introduction

The kidneys are vital organs responsible for filtering blood, removing metabolic waste products, and regulating electrolytes as well as fluid and acid–base balance. [1] Structurally, they are divided into cortex and medulla, which contain the functional units of the kidney, the nephrons. [2] A wide range of diseases can affect the kidneys and their function, including tumors, stones, infections, inflammatory conditions, and cystic diseases. Chronic kidney disease (CKD) is reached when kidney damage and/or a reduced glomerular filtration rate (GFR) have been present for at least three months. CKD is highly frequent worldwide, can lead to cardiovascular diseases, and ranks among the leading causes of death. [3, 4]

Medical imaging of the kidneys, such as computed tomography (CT), plays an important role in detecting abnormalities, determining disease progression, and evaluating potential kidney donors. In particular, quantitative analysis of kidney structure can provide valuable indicators of renal health. Previous studies have shown that cortical shrinkage correlates with both aging and CKD, reduced medullary volume is associated with an increased risk of graft failure after transplantation, and kidney donors with below-average nephron number – estimated from cortical volume – have a higher risk of developing CKD within ten years after donation. [5]

As the global burden of CKD and related diseases continues to increase, obtaining information about kidney anatomy is important for both treatment and research. [4] However, manual annotation of kidney substructures is time-consuming and impractical for large-scale studies or routine clinical workflows. Therefore, there is a need for automated, AI-based methods for fast and detailed segmentation of the kidneys and their substructures, such as cortex, medulla, and cysts.

The aim of this project is to implement and evaluate an AI-driven segmentation framework for cortex, medulla, and cyst segmentation from contrast enhanced CT scans, and to investigate whether the resulting volume measurements can be meaningfully associated with estimated glomerular filtration rate (eGFR).

CHAPTER 2

Clinical Background

This chapter provides an overview of the clinically relevant foundations for kidney segmentation. Section 2.1 gives a review of normal renal anatomy and physiology, while section 2.2 outlines common abnormalities and kidney-related diseases that influence organ appearance, structure and performance. Finally, 2.3 introduces key aspects of abdominal and renal computed tomography (CT) imaging, highlighting acquisition considerations that impact segmentation tasks.

2.1 Renal Anatomy and Physiology

The kidneys are part of the urinary tract, which consists of the upper urinary tract: the kidneys, and the lower urinary tract: the ureters, bladder, and urethra. [6] The overall anatomy of the urinary tract, including internal structures of the kidney, can be seen in figure 2.1.

The kidneys are bean-shaped organs, which each have a weight of approximately 130-150 g, are 10-12 cm long, 5-7 cm wide and 3-4 cm thick. Each kidney is protected by a fibromuscular capsule, which is surrounded by a layer of perirenal fat, connective tissue called renal fascia, and another layer of pararenal fat. [7] Usually a human has two kidneys placed in the posterior abdominal wall, one on each side of the spine, adjacent to the upper lumbar vertebrae. The left kidney is placed slightly higher than the right, because of the position of the liver. [7] However, the body can function with only one kidney, and generally if one of the kidneys is damaged, it will have minimal impact on the overall function and performance of the renal system as long as the other kidney is normal. Nevertheless, abnormalities in a single kidney may still increase vulnerability to infection. [6]

The primary function of the kidneys is to: (1) maintain the body composition by regulating the volume of fluid in the body, and regulating the molarity, electrolyte content and concentration, and the acidity of the fluid; (2) excrete waste products from metabolism and foreign substances; and (3) produce and secrete certain en-

zymes and hormones. [1]

These functions are achieved by filtration of the blood and production of urine. When the urine is produced by the kidney it is drained into the renal pelvis, which is the dilated part of the ureter. From here the collected urine is transported into the bladder through the ureter by peristaltic action. Lastly, the urine is excreted through the urethra. [6]

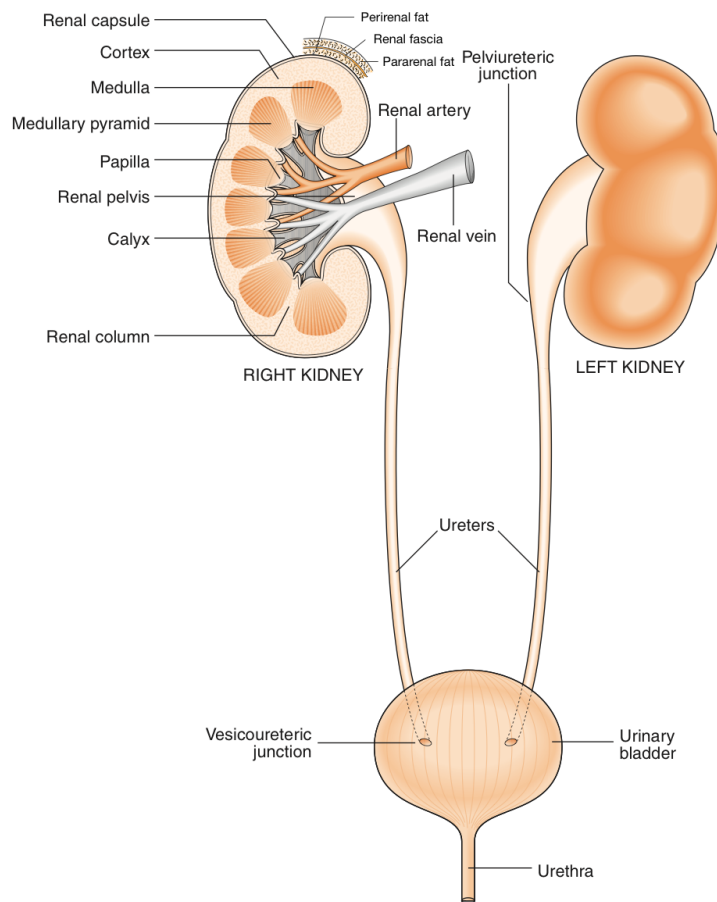


Figure 2.1: Principal anatomical components of the urinary tract, including features seen on a cut surface of the kidney. (Modified from Fields, Harris, and Pollock: *The Renal System* [6]).

We will now take a closer look on what is happening inside the kidney during urine production. In figure 2.1, a cross section can be seen in the right kidney. The left kidney has similar internal structures. An indentation can be seen in the center of the kidney. This is where nerves, the renal pelvis, and the renal artery and vein pass through. The renal artery transports the unfiltered, oxygenized blood into the kidney and the renal vein transports the filtered, deoxygenized blood out again. The renal tissue is divided into two different regions. The outermost region is called the cortex and the inner region is called the medulla. Cortex and medulla are built up by nephrons, which are the functional units of the renal tissue, blood,

vessels, lymphatics and nerves. Medulla is divided into pyramids, which are triangular pieces of tissue. The bases of a pyramid is placed at the border between cortex and outer medulla, and the tips point into the renal papilla. The papilla is placed within the calyx, which is where the urine is collected and passed into the pelvis. [2]

A human has around 1.2 million nephrons. The composition of these can be seen in figure 2.2. A nephron is a hollow tube consisting of a renal corpuscle and a long tubule. The renal corpuscle is composed of glomerulus, a cluster of blood capillaries held together by a cell stack called mesangium and surrounded by cells called podocytes, and Bowman's capsule. The tubule is divided into proximal tubule, loop of Henle, and distal tubule. The distal tubule ends in the collecting duct system, which can be shared by several nephrons. [2]

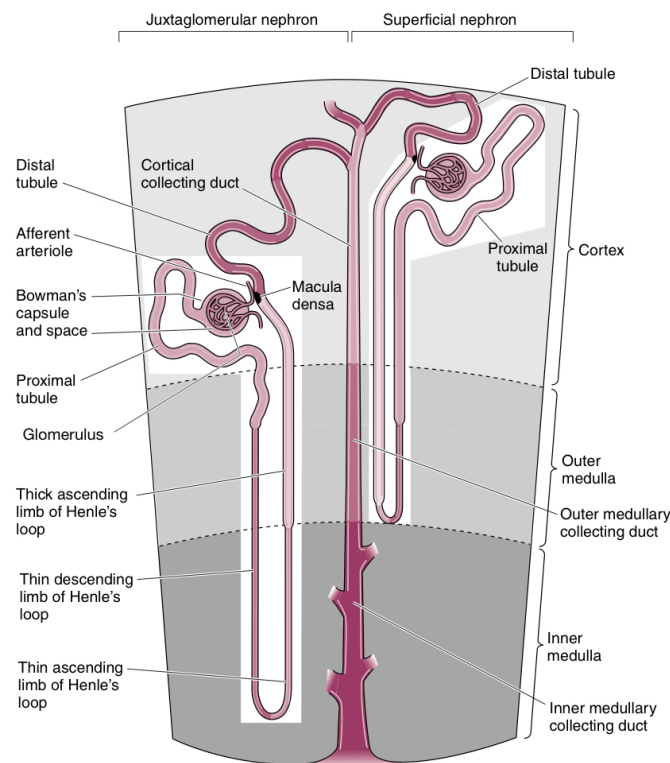


Figure 2.2: Diagram of a juxtaglomerular nephron (left) and a superficial nephron (right), which are two types of nephrons. They differentiate by how deep into the medulla the loop of Henle reaches and how close to the cortex–medulla boundary the glomerulus is placed. (Modified from Koeppen and Stanton: *Renal Physiology* [2]).

Although the kidneys are relatively small organs, they receive 20% of the cardiac output through the renal artery, making them the organs with the highest tissue-specific blood flow. When the blood reaches the glomerulus, it is filtered through the glomerular filtration barrier, which allows water and small dissolved solutes (like ions and metabolic waste products) to pass into the nephron, but blocks

larger molecules like proteins and blood cells. The kidney's performance can be measured through the glomerular filtration rate (GFR). [1]. Next, the filtrate enters the tubule, where the purpose is to reabsorb and secrete fluids and electrolytes in order to adjust the content of the urine optimally, with the aim of maintaining the the body composition as described above. [6] The different segments of the tubule are made up of different cells with specific transport functions. Lastly, the filtrate reaches the collecting duct, which is composed by principal cells and intercalated cells. These cell types also have specific transport functions. For instance, principal cells are important for absorption of NaCl and secretion of K^+ , and the intercalated cells are important for acid-base regulation. When the filtrate has reached the inner medullary collecting duct it ends at the papilla as urine, ready to be transported out of the kidney. [2]

Production and secretion of hormones and enzymes happen different places in the kidney. Renin, which is an enzyme that catalyzes formation of angiotensinogen (a peptide that contributes to regulation of salt balance and blood-pressure), is produced in the glomerulus. The protein erythropoietin is produced by specific cells in the cortex and helps maturing red blood cells in the bone marrow. Another example is the most active form of vitamin D₃, which is important for regulation of calcium and phosphate balance and produced in the proximal tubule cells. [1]

Generally, kidney function influences many aspects of the body. For example, the regulation of fluid volume and ion concentrations is critical for maintaining optimal plasma volume, on which cardiac output and blood pressure depend. In addition, enzyme activity, cell membrane potential, and membrane excitability all rely on the proper balance of specific ions. [1]

2.2 Diseases Affecting the Kidneys

Several abnormalities and diseases can arise in the kidney. Common structural abnormalities are kidney stones, renal tumors, and cysts, for instance caused by polycystic kidney disease (PKD). PKD is a genetic disorder in which fluid-filled cysts grow in the kidneys, expanding and changing the shape of the kidneys. [8] Other conditions are pyelonephritis, which is generally a bacterial infection that has spread to the kidney, and glomerulonephritis, which causes inflammation in the glomeruli. [9] [10] These abnormalities and diseases can damage the kidney and affect kidney function.

Chronic kidney disease (CKD) is defined as evidence of either kidney damage or a GFR of less than 60 mL/min/1.73 m², or both, lasting for 3 months or longer. The condition reflects a gradual decline in kidney function that, if progressive, can lead to end-stage kidney disease (ESKD), where the GFR is less than 15 mL/min/1.73 m². ESKD requires dialysis or transplantation. [3]

The reasons for developing CKD vary, but the most common causes are diabetes and hypertension. Type 2 diabetes is responsible for around 30–50% of the cases, type 1 diabetes for around 3.9% of the cases, and hypertension for around 27.2% of the cases. Other relatively common causes are glomerulonephritis and cystic diseases such as PKD. [11]

CKD can lead to several complications, with cardiovascular diseases such as heart attacks and strokes being among the most serious. People with CKD face a significantly higher risk of developing these conditions, and the risk increases as CKD progresses. In fact, people with CKD are five to ten times more likely to die than to progress to ESKD, mainly due to cardiovascular diseases. [3]

Reduced kidney function is a large problem world wide. Around 850 million people have CKD, acute kidney injury (AKI), which is a sudden and often reversible loss of kidney function [12], or are on renal replacement therapy, which includes dialysis and kidney transplantation. This is about 10% of the population. In 2016, CKD was ranked as the 13th leading cause of death worldwide, and by 2040 it is projected to rise to the 5th leading cause of years of life lost. [4]

2.3 Computed Tomography

CT is an abbreviation for computed tomography. In order to make a CT scan of a patient, the patient is placed on a table within a donut-shaped CT-scanner, while the scanner rotates around the body in an axial plane emitting X-rays. X-ray detectors receive the X-ray beam and measure how much of the X-ray energy was absorbed in the body. By moving the table through the scanner, while the scanner rotates, the X-ray absorbed in each voxel in each slice of the body can be calculated by a computer, and a corresponding numerical value can be assigned to the voxel. [13][14]

X-ray absorption in a known slice thickness varies in proportion to the density of the tissue examined. This means each voxel can be associated to a linear attenuation coefficient, μ , which can be converted into Hounsfield Units (HU) by using the linear attenuation coefficient of water, μ_{water} . Conversion into HU is given by

$$\text{CT}_{\text{num}}(\mu) = 1000 \cdot \frac{\mu - \mu_{\text{water}}}{\mu_{\text{water}}}, \quad (2.3.1)$$

where CT_{num} indicates the CT-number, i.e. grayscaled pixel-value given in HU. From equation (2.3.1) we see that the Hounsfield scale is constructed such that the HU value of water is 0. Other HU values for air and different types of tissues can be seen in figure 2.3. [15]

In a CT scan of the abdomen, different organs can be seen. Among these are the kidneys, liver, spleen, gallbladder, pancreas, stomach, colon, small bowel, duode-

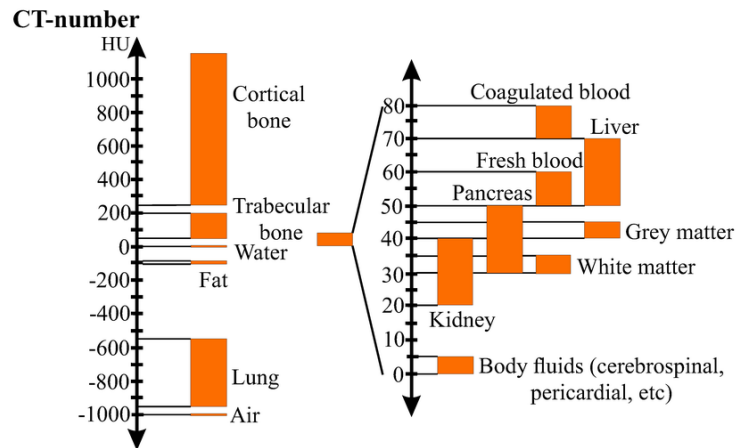


Figure 2.3: Hounsfield scale for different kinds of tissue. (Original from Amadeus Alcantara et. al: *Patient-Specific Bone Multiscale Modelling, Fracture Simulation and Risk Analysis — A Survey* [15]).

num, urinary bladder, and adrenal glands. Blood vessels such as the inferior vena cava (IVC), aorta, and the portal and splenic vein can also be seen. Figure 2.4 shows two slices in axial and coronal view, respectively, where different parts of these organs and vessels can be seen.

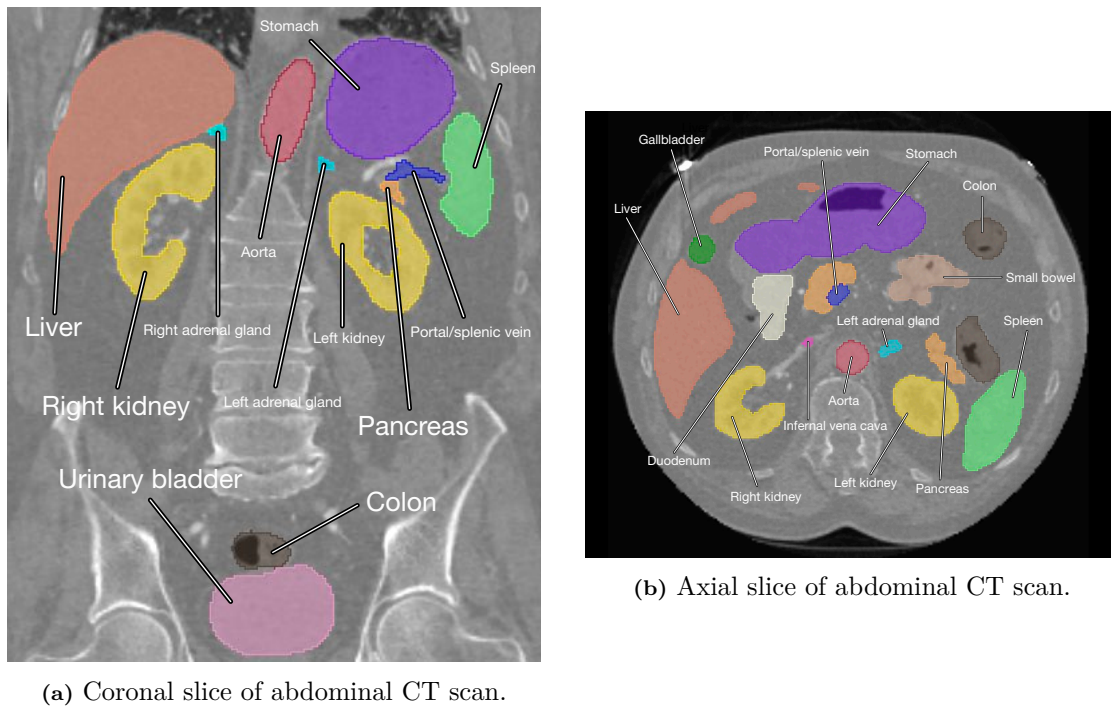


Figure 2.4: Coronal and axial view of CT scan showing the abdominal area. The CT scan and annotations of organs and vessels are from a sample of the TotalSegmentator dataset. The scan is contrast-enhanced. Labels have been written on top of the figure.

As illustrated in figure 2.3 the HU values of soft tissues like liver, kidney, and pancreas tissue lie within a rather small range. This means that it can be hard to see a difference between tissue types in the abdomen on a CT. For instance, it is very hard to differentiate between cortex and medulla in the kidney.

However, CT scans can be contrast enhanced in order to improve visualization. A contrast agent is injected in a vein located in the extremities, e.g. in the arm, and travels through the blood vessels. When it reaches the vessels in an organ this organ will light up (enhance) in a CT scan if the scan is performed before the contrast agent has left the organ again. The different phases of CT refer to this timing of scan after injection. The most used phases are the arterial, late arterial, portal venous, urographic and delayed phase, where the scan is performed 15 seconds, 30–40 seconds, 70 seconds, 90 seconds and 5–20 minutes after injection, respectively. [16]

The kidneys receive 25% of the cardiac output as described earlier, and this means the kidneys enhance quickly after injection of the contrast agent. The blood first reaches cortex, then medulla. Therefore, if the arterial phase is selected, we will be able to see enhancement of the kidneys, specifically cortex, creating a visual separation of cortex and medulla as seen in figure 2.5. This phase, where cortex is seen to enhance more than medulla, is also called the corticomedullary phase. [17] [18]

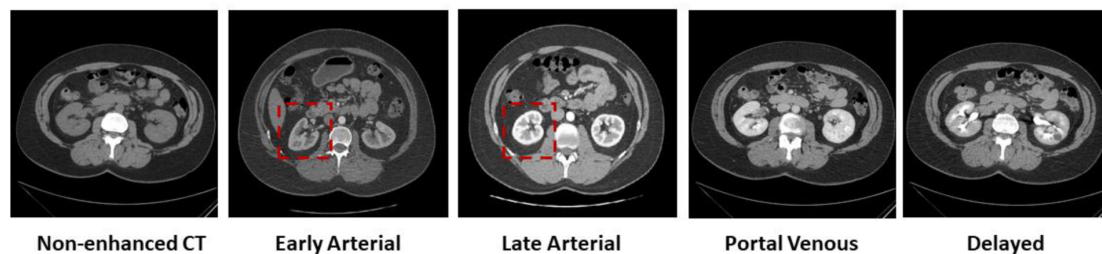


Figure 2.5: Contrast enhancement phases in CT images, including non-enhanced CT, early arterial phase, late arterial phase, portal venous phase and delayed phase CT. In the arterial phase a clear difference in contrast between cortex and medulla in the kidney can be seen. The right kidney is marked with red boxes. (Original from Yucheng Tang et. al: *Renal Cortex, Medulla and Pelvicaliceal System Segmentation on Arterial Phase CT Images with Random Patch-based Networks* [17]).

Although contrast enhancement is necessary in order to distinguish between cortex and medulla, non-enhanced CT can be useful for other purposes. For instance, if the goal is to detect kidney stones. [16]

CHAPTER 3

Technical Background and Methods

This chapter introduces the theoretical foundation on which the methods and concepts mentioned in this bachelor project are built. In section 3.1 neural networks (NNs) are introduced, specifically multi-layer perceptrons (MLPs) and convolutional neural networks (CNNs). Section 3.2 explains the machine learning method Gradient Tree Boosting, and in section 3.3 some relevant evaluation metrics used in classification and image segmentation are discussed. Finally, section 3.4 presents three different statistical hypothesis tests.

3.1 Neural Networks

Neural networks (NNs) is a class of computational models made up of layers of connected units (neurons) that take inspiration from the way biological neurons process information. In the following, two types of neural networks are explained.

A **multi-layer perceptron** (MLP) is a type of fully connected neural network composed of a sequence of layers: an input layer, one or more hidden layers, and an output layer. Each neuron in a hidden or output layer is connected to all neurons in the previous layer, and each of these connections has an associated learnable weight. Neurons in these layers also have associated bias terms. The input layer does not perform any computation, but presents the input features to the network. Computation begins in the hidden layers: each neuron receives numerical inputs from the previous layer, forms a weighted sum of these inputs, adds a bias, applies a nonlinear activation function, and produces an output. This output then serves as input to neurons in the next layer. Assume that layer $l - 1$ contains K neurons. For any hidden or output layer l , the output of neuron j is given by

$$a_j^{(l)} = \sigma(z_j^{(l)}), \quad z_j^{(l)} := \sum_{k=1}^K w_{jk}^{(l)} a_k^{(l-1)} + b_j^{(l)},$$

where $w_{jk}^{(l)}$ is the weight on the connection from neuron k in layer $l - 1$ to neuron j in layer l , $b_j^{(l)}$ is the bias associated with neuron j in layer l , and σ is a non-linear activation function.

Different **activation functions** can be applied. Frequently used ones are the rectified linear unit (ReLU) defined by $x \mapsto \max\{0, x\}$, and the modification named leaky ReLU, which is defined by $x \mapsto \max\{\alpha x, x\}$, where α is a small positive constant. Leaky ReLU prevents $a_j^{(l)}$ and $\sigma'(z_j^{(l)})$ from being zero for negative $z_j^{(l)}$. Another commonly used activation function is the sigmoid, defined by $x \mapsto 1/(1 + e^{-x})$. It is often used in the output layer for tasks such as binary classification, because its output can be interpreted as a probability. For multi-class problems we can use softmax, defined by $z_j \mapsto e^{z_j} / \sum_{c=1}^C e^{z_c}$ for $j = 1, \dots, C$. This activation is applied across all C neurons in the output layer, ensuring that the outputs form a valid probability distribution that sums to 1, where C is the number of classes.

3.1.1 Convolutional Neural Networks

Convolutional neural networks (CNNs) are designed for 2D or 3D data such as images. They use local receptive fields, shared weights, and pooling to efficiently detect patterns. In addition to fully connected layers, CNNs include convolution and pooling layers.

In a convolution layer, an $n \times n$ kernel, in which the values are the weights, slides over the image and computes a weighted sum of the overlapping pixels and adds a bias, followed by an activation function. In this way, neuron interacts only with a small neighbourhood of the input, and the same weights are applied everywhere, allowing the network to detect features regardless of position. The number of input channels sets the depth of each kernel. The number of kernels determines the number of output channels. The stride of the kernel can be adjusted to control downsampling, and padding can be added to preserve spatial dimensions. An example of a convolution can be seen in figure 3.1.

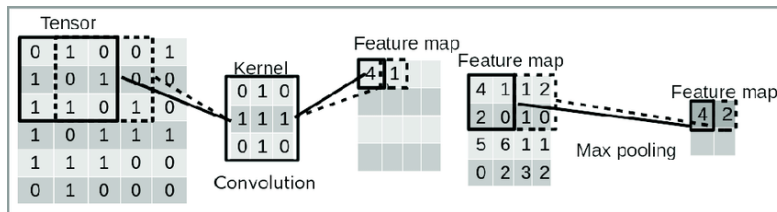


Figure 3.1: Example of a 3×3 convolution with stride 1 and a 2×2 max pooling with stride 2. (Original from Jan Schäfer et al.: *Measuring Particle Size Distributions in Multiphase Flows Using a Convolutional Neural Network* [19]).

In a pooling layer—typically placed after convolutional layers—the feature maps are summarized and downsampled. A common choice is max pooling, where each

$n \times n$ region is replaced by its maximum value. For example, applying 2×2 max pooling halves the spatial dimensions of the feature maps. Pooling operates independently on each channel.

Transposed convolution can be used to increase spatial resolution. It conceptually works by expanding the input, effectively inserting gaps between pixels/voxels according to the stride, and then applying a convolutional kernel to fill in these gaps, allowing the network to learn how to upsample the feature maps. An example in 2D can be seen in figure 3.2. A 2×2 transposed convolution with stride 2 (or $2 \times 2 \times 2$ in 3D) is often used, since this doubles the spatial dimensions.

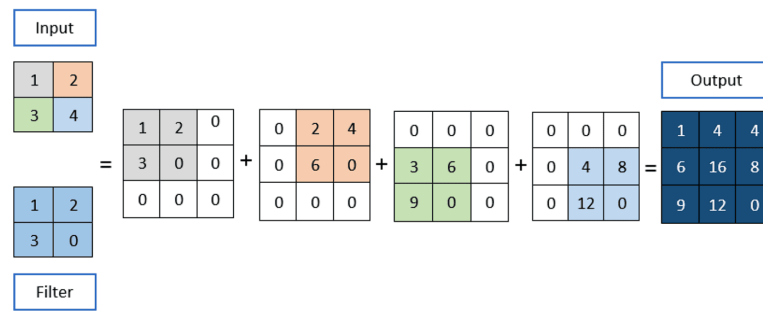


Figure 3.2: Example of a 2×2 transposed convolution with stride 1 on a 2×2 image. (Original from Md Afif Al Mamun and Imamul Kadir: "an-Eye: Safe navigation in footpath for visually impaired using computer vision techniques" [20]).

U-Net is a specific type of CNN architecture designed for image segmentation, originally developed for biomedical images. It has a symmetric encoder-decoder structure shaped like a "U", hence the name. In the encoder, convolutions and downsampling are performed to reduce spatial dimensions while increasing the number of channels, allowing the network to learn important features. In the decoder, the spatial and channel dimensions are gradually restored through upsampling and convolutions. Skip connections concatenate features from the encoder with those in the decoder, helping the network combine an understanding of the overall image with precise spatial details for accurate segmentation. In the original 2D and 3D versions [21][22], valid convolutions without padding were used. As a result, the encoder feature maps had to be cropped in the skip connections to match the spatial dimensions in the decoder. Because the final segmentation mask would otherwise be significantly smaller than the input, mirroring was applied to the images so border pixels could be segmented correctly. A version of a 3D U-Net can be seen in Figure 6.1 in Chapter 6.

3.1.2 Training a Neural Network

In order to train a neural network, a loss function is optimized with respect to the kernel weights. Typical loss functions for image segmentation, which were used to train the models applied in this project, are the soft Dice loss, Tversky loss, and

Cross-entropy (CE) loss. Cross-entropy loss is defined as

$$CE(y, p) = - \sum_{n=1}^N \log(y_n \cdot p_n),$$

where y_n is the true one hot encoding of classes for pixel n , p_n is the vector of predicted class probabilities for pixel n , and $\cdot$ is the dot product. [23] By definition of CE, each pixel contributes equally to the loss. This can be challenging when dealing with small classes if class weights aren't used. In contrast, each class contributes equally, regardless of how many pixels it has, when using the soft Dice loss. This loss is based on the Dice coefficient, which measures the overlap between predicted and true masks, see section 3.3, but ensures differentiability by using the probability outputs of the model. It is defined as

$$\mathcal{L}_{Dice}(y, p) = 1 - \frac{1}{C} \sum_{c=0}^{C-1} \frac{2 \sum_{n=1}^N y_n^c p_n^c}{\sum_{n=1}^N (y_n^c + p_n^c)},$$

where C is the number of classes, N is the number of pixels, y_n^c is an indicator variable being 1 if pixel n belongs to class c and 0 otherwise, and p_n^c is the model probability output corresponding to pixel n belonging to class c . [23] The Tversky function is similar, but based on the Tversky index, where false positives and false negatives are weighted by parameters α and β , respectively. This loss is defined

$$\mathcal{L}_{Tversky}(y, p) = 1 - \frac{1}{C} \sum_{c=0}^{C-1} \frac{\sum_{n=1}^N y_n^c p_n^c}{\sum_{n=1}^N (y_n^c p_n^c + \alpha p_n^c (1 - y_n^c) + \beta y_n^c (1 - p_n^c))},$$

Note that if $\alpha = \beta = \frac{1}{2}$, this reduces to soft Dice loss. [23]

In order to minimize the loss functions, backpropagation is applied to compute the gradients of the loss with respect to the network weights, and weights are updated using optimizers. The learning rate controls the step size. In order to prevent overfitting many different things can be done. This includes the use of dropout, for instance spacial dropout in CCNs, where the activations in entire channels are randomly set to zero to prevent overfitting while preserving spatial structure within feature maps.

3.2 Gradient Tree Boosting

The machine learning model Gradient Tree Boosting is a tree ensemble model that uses a series of regression trees to make a prediction. This section is heavily based on derivations in the XGBoost article by Chen and Guestrin (2016) [24]. We start by mathematically defining the concept of a regression tree.

3.2.1 Regression trees

Consider a data set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, with features $\mathbf{x}_i \in \mathbb{R}^m$, targets $y_i \in \mathbb{R}$, and cardinality $|\mathcal{D}| = n$. Furthermore let $\mathcal{F}$ be the space of regression trees. Now, for a regression tree $f_k \in \mathcal{F}$, with the set of leaves T_k , we define the tree as a function $f_k : \mathbb{R}^m \rightarrow \mathbb{R}$ by

$$f_k(\mathbf{x}) = w_{q_k(\mathbf{x})}, \quad (3.2.1)$$

where $q_k : \mathbb{R}^m \rightarrow T_k$, $q_k(\mathbf{x}) = t$, is a function that assigns each input $\mathbf{x}$ to the leaf $t \in T_k$ reached by traversing the tree according to the decision rules, and $w_{q_k(\mathbf{x})}$ is the continuous score (weight) on leaf t . Note that we denote the vector of all leaf scores in the tree by $w \in \mathbb{R}^T$, where $T = |T_k|$, so $w_{q_k(\mathbf{x})}$ is an element in w . The obtained weight $w_{q_k(\mathbf{x})}$ defines the prediction obtained by the tree f_k given the input $\mathbf{x}$.

3.2.2 Tree Boosting

We can now describe how an ensemble tree model is used to predict an output $\hat{y}_i$ from an input $\mathbf{x}_i$. Given a series of K regression trees $\{f_k\}_{k=1}^K$, the predicted output is

$$\hat{y}_i = \phi(\mathbf{x}) = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F}. \quad (3.2.2)$$

To learn the functions f_k , i.e. the structure of the trees, an objective function is minimized. This function has two terms: a term involving a convex, differentiable loss function l and a regularization term, and is defined

$$\mathcal{L}(\phi) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k), \quad (3.2.3)$$

where $\Omega : \mathcal{F} \rightarrow \mathbb{R}$ is a function that describes the complexity of a tree. In the XGBoost implementation they have defined the complexity as

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2, \quad (3.2.4)$$

where $T = |T_k|$, and γ and λ are non-negative regularization parameters penalizing the size of the tree and the leaf weights, respectively. Note that $\frac{1}{2} \lambda \|w\|^2$ is the L2 regularization term, which shrinks the sizes of the weights.

3.2.3 Gradient Tree Boosting

Instead of learning all trees at once to optimize $\mathcal{L}(\phi)$, the trees f_k are trained additively. Starting with one tree we add a new tree at each iteration such that the tree at iteration k is trained to learn the negative gradient scaled by the inverse hessian at iteration $k-1$. To understand this, we first define the predictions additively. If we initialize $\hat{y}_i^{(0)} = 0$, the predictions in iterations 1 through j given

$\mathbf{x}_i$ are

$$\begin{aligned}\hat{y}_i^{(1)} &= \phi_1(\mathbf{x}_i) = f_1(\mathbf{x}_i) = \hat{y}_i^{(0)} + f_1(\mathbf{x}_i), \\ \hat{y}_i^{(2)} &= \phi_2(\mathbf{x}_i) = f_1(\mathbf{x}_i) + f_2(\mathbf{x}_i) = \hat{y}_i^{(1)} + f_2(\mathbf{x}_i), \\ &\vdots \\ \hat{y}_i^{(j)} &= \phi_j(\mathbf{x}_i) = \sum_{k=1}^j f_k(\mathbf{x}_i) = \hat{y}_i^{(j-1)} + f_j(\mathbf{x}_i).\end{aligned}$$

Consequently, the objective function at iteration j , that tree f_j is trained to minimize, is

$$\mathcal{L}(\phi_j) = \sum_{i=1}^n l(y_i, \hat{y}_i^{(j)}) + \Omega(f_j) = \sum_{i=1}^n l(y_i, \hat{y}_i^{(j-1)} + f_j(\mathbf{x}_i)) + \Omega(f_j) \quad (3.2.5)$$

We can approximate eq. (3.2.5) using second order Taylor expansion of the loss function $l(y_i, \hat{y}_i^{(j-1)} + f_j(\mathbf{x}_i))$ around $\hat{y}_i^{(j-1)}$. This yields

$$\mathcal{L}(\phi_j) \simeq \sum_{i=1}^n \left(l(y_i, \hat{y}_i^{(j-1)}) + g_i f_j(\mathbf{x}_i) + \frac{1}{2} h_i f_j(\mathbf{x}_i)^2 \right) + \Omega(f_j), \quad (3.2.6)$$

where $g_i = \frac{\partial l(y_i, \hat{y}_i^{(j-1)})}{\partial \hat{y}_i^{(j-1)}}$ is the gradient and $h_i = \frac{\partial^2 l(y_i, \hat{y}_i^{(j-1)})}{\partial (\hat{y}_i^{(j-1)})^2}$ is the hessian. Minimizing (3.2.6) is equivalent to minimizing

$$\tilde{\mathcal{L}}(\phi_j) = \sum_{i=1}^n \left(g_i f_j(\mathbf{x}_i) + \frac{1}{2} h_i f_j(\mathbf{x}_i)^2 \right) + \Omega(f_j) \quad (3.2.7)$$

$$= \sum_{i=1}^n \left(g_i f_j(\mathbf{x}_i) + \frac{1}{2} h_i f_j(\mathbf{x}_i)^2 \right) + \gamma T + \frac{1}{2} \lambda \sum_{t=1}^T w_t^2 \quad (3.2.8)$$

$$= \sum_{t=1}^T \left(\left(\sum_{i \in I_t} g_i \right) w_t + \frac{1}{2} \left(\sum_{i \in I_t} h_i + \lambda \right) w_t^2 \right) + \gamma T, \quad (3.2.9)$$

where $I_t = \{i \mid q_j(\mathbf{x}_i) = t\}$ is the set of indices i for which $\mathbf{x}_i$ is assigned to leaf t . Recall that T is the total number of leaves in the tree f_j . If we differentiate $\tilde{\mathcal{L}}(\phi_j)$ with respect to $f_j(\mathbf{x}_i)$, we get

$$\frac{\partial \tilde{\mathcal{L}}(\phi_j)}{\partial f_j(\mathbf{x}_i)} = 0 \Leftrightarrow f_j(\mathbf{x}_i) = -\frac{g_i}{h_i}.$$

Hence, if each $f_j(\mathbf{x}_i)$ were optimized independently - rather than through the tree structure - the optimal update for each data point would be $f_j(\mathbf{x}_i) = -\frac{g_i}{h_i}$. In practice, however, we optimize the tree f_j itself, which can only assign a single constant value per leaf. Therefore, the tree f_j effectively learns a piecewise-constant ap-

proximation of the ideal values $-\frac{g_i}{h_i}$: all points that fall into the same leaf share the same prediction.

Arriving at the objective in (3.2.9) we see that when q_j , so the tree structure, is fixed, we obtain the optimal weight on leaf t

$$w_t^* = -\frac{\sum_{i \in I_t} g_i}{\sum_{i \in I_t} h_i + \lambda}, \quad (3.2.10)$$

by solving $\frac{d}{dw_t} \tilde{\mathcal{L}}(\phi_j) = 0$, for which the optimal value of the objective is

$$\tilde{\mathcal{L}}(\phi_j, q_j) = -\frac{1}{2} \sum_{t=1}^T \frac{(\sum_{i \in I_t} g_i)^2}{\sum_{i \in I_t} h_i + \lambda} + \gamma T. \quad (3.2.11)$$

The second argument of the objective indicates that the function q_j was fixed. Now, the question left is how to split the nodes in the tree, i.e. how to define $q_j(\mathbf{x}_i)$ for $i = 1, \dots, n$. Instead of investigating all possible tree structures, greedy algorithms are used to decide how a node should be split, based on the gain. Considering a set I of indices assigned to a leaf, we let I_L and I_R denote the index set of the resulting left and right node after the split, i.e. $I = I_L \cup I_R$, $I_L \cap I_R = \emptyset$. First, define the local contribution to the objective, of a leaf with index set I , as

$$\tilde{\mathcal{L}}(I) = -\frac{1}{2} \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} + \gamma. \quad (3.2.12)$$

The gain by splitting the node is then

$$Gain = \tilde{\mathcal{L}}(I) - (\tilde{\mathcal{L}}(I_L) + \tilde{\mathcal{L}}(I_R)) \quad (3.2.13)$$

$$= \frac{1}{2} \left(\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right) - \gamma. \quad (3.2.14)$$

If the gain is positive, the objective is reduced, meaning the split is favourable. At each step, a greedy algorithm evaluates all possible splits of a leaf I and chooses the one with the highest positive gain. A split is only accepted if $Gain > 0$.

In the XGBoost implementation of Gradient Tree Boosting there are different hyperparameters available to tune in order to prevent overfitting for example. Besides the regularization parameters, a learning rate $\eta \in (0, 1]$ can be modified such that the prediction will be a weighted sum $\hat{y}_i^{(k)} = \hat{y}_i^{(k-1)} + \eta f_k(\mathbf{x}_i)$. That is, the learning rate scales the contribution of each new tree, controlling the step size of the additive updates. Other parameters include for instance the maximum depth of a tree, maximum number of leaves etc.

Note that when using the squared error, i.e. $l(y_i, \hat{y}_i^{(j-1)}) = \frac{1}{2}(y_i - \hat{y}_i^{(j-1)})^2$, as the

loss function, the problem simplifies, as then $g_i = (\hat{y}_i^{(j-1)} - y_i)$, and $h_i = 1$. In this case, a tree f_j is trained to predict exactly the residuals associated with the previous prediction $\hat{y}_i^{(j-1)}$.

3.3 Evaluation Metrics

The F1 score, an evaluation metric used in classification, is defined as the harmonic mean of precision and recall. These quantities are given by

$$Precision := \frac{TP}{TP + FP}, \quad Recall := \frac{TP}{TP + FN},$$

where TP is the number of true positives, FP is the number of false positives, and FN is the number of false negatives. The F1 score is therefore

$$F1 := \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}.$$

Since precision and recall each lie within $[0,1]$, the F1 score also lies within this interval. A higher F1 score corresponds to better classifier performance. The Dice similarity coefficient (DSC) is a similarity measure between two sets A and B , defined as

$$Dice(A, B) := \frac{2|A \cap B|}{|A| + |B|}.$$

Like the F1 score, the DSC is bounded between 0 and 1 and attains its maximum value of 1 when the sets perfectly overlap. The DSC is used as an evaluation metric in binary image segmentation. If we define the sets Y and $\hat{Y}$ as the sets of pixel locations in the ground truth mask and predicted mask, respectively, where the pixel/voxel value is 1, then $|Y| = TP + FN$ and $|\hat{Y}| = TP + FP$, so $|Y \cap \hat{Y}| = TP$, and then

$$Dice(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}.$$

Therefore, in the context of pixel-level binary classification, i.e. in binary segmentation, the F1 score and the Dice score are mathematically identical.

3.4 Statistical Hypothesis Testing

The three different statistical tests used for hypothesis testing in this project are summarized below.

The **Shapiro–Wilk Test** is a statistical test evaluating whether a random sample is drawn from a population with normal distribution. The null hypothesis and

alternative hypothesis are

H_0 : the sample is drawn from a population with normal distribution,

H_1 : the sample is not drawn from a normal distribution.

The Shapiro–Wilk test works best for smaller sample sizes and the test statistic is

$$W = \frac{\left(\sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} a_{i,n} (X_{(n-i+1)} - X_{(i)}) \right)^2}{\sum_{i=1}^n (X_i - \bar{X})^2},$$

where $a_{i,n}$ are constants depending on n , $\bar{X}$ is the sample mean, and $X_{(i)}$ are the i th order statistics. The critical value w_α corresponding to significance level α satisfies $P(W \leq w_\alpha) = \alpha$, and in order for H_0 to be rejected on a significance level α , the observed value of the test statistic must be in the critical region, that is, $w_{obs} < w_\alpha$. Equivalently, H_0 is rejected if $p < \alpha$, where $p = P(W \leq w_{obs})$. Otherwise, it is accepted. [25]

The two-sided independent two-sample **Welch’s t-test** is used to compare the means of two independent populations with normal distributions $\mathcal{N}(\mu_1, \sigma_1^2)$ and $\mathcal{N}(\mu_2, \sigma_2^2)$, with possibly unequal variances. Defining the null hypothesis and alternative hypothesis

$$H_0 : \mu_1 = \mu_2,$$

$$H_1 : \mu_1 \neq \mu_2,$$

we get the following test statistic, which approximately follows a t -distribution with $\nu = \left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2 \cdot \left(\frac{1}{n_1-1} \left(\frac{s_1^2}{n_1} \right)^2 + \frac{1}{n_2-1} \left(\frac{s_2^2}{n_2} \right)^2 \right)^{-1}$ degrees of freedom

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Here n_i are the sample sizes, $\bar{X}_i$ are the sample means, and S_i^2 are the sample variances. If the observed test statistic satisfies $|t_{obs}| > t_{1-\alpha/2}$, where the critical value satisfies $P(T \leq t_{1-\alpha/2}) = 1 - \alpha/2$, then H_0 is rejected on a significance level α . Equivalently, if $p < \alpha$, where $p = 2P(T > |t_{obs}|)$, then H_0 is rejected. Otherwise, it is accepted. [26]

Finally, the two-sided independent two-sample **Mann–Whitney U-test** is a non-parametric test used to investigate whether two random samples come from the same population. This is done by determining whether observations in one sample tend to be larger than observations in the other. The null hypothesis and

alternative hypothesis can be defined as

$$H_0 : P(X > Y) = \frac{1}{2},$$

$$H_1 : P(X > Y) \neq \frac{1}{2},$$

where X is a random variable from one sample and Y is a random variable from the other. To calculate the test statistic, the observations from both samples X and Y are first pooled into a single combined dataset. Then the combined observations are ranked from lowest to highest, and the sum of the ranks assigned to the original X observations (denoted R_x) and the sum of the ranks assigned to the original y observations (denoted R_y) are computed. Denoting the sample sizes n_x and n_y , respectively, the test statistic is $U = \min\{U_x, U_y\}$, where

$$U_x = n_x n_y + \frac{n_x(n_x + 1)}{2} - R_x, \quad U_y = n_x n_y + \frac{n_y(n_y + 1)}{2} - R_y.$$

It can be shown that if n_x and n_y are sufficiently large (> 8) a normal approximation can be used. In this case $U \stackrel{a}{\sim} \mathcal{N}(\mu_U, \sigma_U^2)$, where $\mu_U = (n_x n_y)/2$ and $\sigma_U^2 = (n_x n_y)(n_x + n_y + 1)/12$. In the case of rank ties, the variance is adjusted. The approximately normally distributed test statistic then becomes

$$Z = \frac{U - \mu_U}{\sigma_U} \sim \mathcal{N}(0, 1),$$

and the decision rule is to reject H_0 if $|z_{obs}| > z_{1-\alpha/2}$ or equivalently if $p < \alpha$, where $p = 2P(Z \leq -|z_{obs}|)$. [27]

CHAPTER 4

State-of-the-art

This chapter presents state-of-the-art within kidney segmentation with a focus on cortex-medulla segmentation. The model proposed by Korfiatis et al. (2022) [5] is given special attention, since it plays a important role in this project.

4.1 Kidney Segmentation from CT Scans

Segmentation of the kidneys is an important and clinically relevant task. It supports the diagnosis of renal conditions such as stones and tumors, enables tracking of disease progression (e.g., in polycystic kidney disease), and helps identify suitable donors for transplantation by measuring kidney volume [5]. However, manual segmentation is time-consuming, requires expert knowledge, and has limited scalability, making it impractical.

In the last decade deep learning methods, particularly the 2D and 3D U-Net architecture introduced by Ronneberger et al. (2015) [21] and Çiçek et al. (2016) [22] and their variants, have revolutionized medical image segmentation. The nnU-Net proposed by Isensee et al. (2018) [28], which is a self-configuring framework based on U-Net, and TotalSegmentator introduced by Wasserthal et al. (2023) [29], which is a large model for multi-organ segmentation trained using nnU-Net, report performance among state-of-the-art across diverse datasets and medical image segmentation tasks, including whole kidney segmentation from CT scans.

Although most studies focus on whole kidney segmentation, differentiating between cortex and medulla is also clinically relevant. For instance, an accelerated reduction of cortical volume compared to medullary volume is a characteristic feature of both ageing and CKD. [5] Studies by Tang et al. (2021) [17], Korfiatis et al. (2022) [5], Yu et al. (2023) [30], and Chen et al. (2025) [31] are among state-of-the-art-performance in this area. These studies are described in the sections below.

More recently, the Transformer architecture presented by Vaswani et al. (2017) [32],

which builds on attention mechanisms and was adapted for computer vision with the Vision Transformer (ViT) by Dosovitskiy et al. (2021) [33], has been adopted in medical imaging. Particularly, hybrids between U-Nets and transformers have been used by Yu et al. (2023) [30], and Chen et al. (2025) [31].

4.1.1 Tang et al. (2021)

Tang et al. (2021) [17] propose a coarse-to-fine framework that they call Random Patch Network Fusion (RPFN) to segment the renal cortex, medulla, and pelvicalyceal system from arterial-phase CT scans. The method proceeds in two stages, both based on a traditional 3D U-Net backbone. In the coarse stage they trained a 3D U-Net on CT scans resampled to low resolution volumes in order to segment the whole kidney. In the fine stage they first used the coarse kidney mask to randomly select voxels within the kidney in each CT scan. Then they used a bounding box to extract 3D patches. Another 3D U-Net was trained on the random patches, now at original resolution, to distinguish cortex, medulla, and pelvicalyceal system. Because patches overlap, each voxel can be labeled multiple times. Therefore, patch predictions are fused by majority voting, ensuring each voxel receives a single final label.

The dataset consisted of 45 arterial phase CT scans and manual annotations of cortex, medulla and pelvicalyceal system on both left and right kidney, annotated under expert supervision. The CT scans were without diagnostic codes involving kidney abnormalities. They used 25 of the scans for 5-fold cross validation and 20 scans for testing. They achieved a DSC of 0.84 for cortex segmentation and a DCS of 0.79 for medulla segmentation.

4.1.2 Korfiatis et al. (2022)

Korfiatis et al. (2022) [5] trained a modified 3D U-Net to segment cortex and medulla in the left and right kidneys. The architecture, which can be seen in figure 6.1, consists of an encoder with four downsampling blocks and a decoder with four corresponding upsampling blocks. All convolutions use a kernel size of $3 \times 3 \times 3$, followed by instance normalization and Leaky ReLU. Downsampling is performed by applying stride 2 in the first convolution of each encoder block, while skip connections concatenate feature maps from the encoder to the decoder at matching resolutions. Upsampling is implemented by repeating voxel values along each axis to double the spatial resolution. Several modifications to the original 3D U-Net architecture are incorporated in both the encoder and decoder. In general, zero-padding is used instead of cropping to ensure that the spatial dimensions remain aligned for the skip connections.

Within each encoder block, a residual connection is used: the output of the first convolution is added to the output of the subsequent double convolution before passing to the next block or skip connection. In addition, spatial dropout with

rate 0.3 is applied after the first convolution.

In the decoder, at the highest three resolution levels, the concatenated encoder-decoder features are processed by a $3 \times 3 \times 3$ convolution followed by a $1 \times 1 \times 1$ convolution. The resulting feature map serves two purposes:

1. It is upsampled, then a $3 \times 3 \times 3$ convolution is applied, and then it is added to the feature map of the next higher-resolution level, as is typical in a 3D U-Net architecture.
2. The same feature map is, at the top three resolution levels, passed through a $1 \times 1 \times 1$ convolution to produce a coarse segmentation with five channels, corresponding to the output classes. This can be interpreted as a coarse segmentation at that resolution. Each of these coarse segmentations is upsampled to match the resolution of the next level and added to the segmentation prediction at that level.

Finally, at the full resolution, a softmax activation is applied across the five channels at each voxel location to produce class probabilities and yield the final segmentation mask. The model is trained using the Tversky loss function with $\alpha = 0.3$ and $\beta = 0.7$, which penalizes false negatives more, and optimized with the Adam optimizer.

Predonation scans of living kidney donors with contrast enhancement (angiogram phase, where cortex, but not medulla is radiopaque) of living kidney donors were used to train test and validate the model. Ground truth was obtained by expert manual segmentation. Out of 1930 CT scans from Mayo Clinic in Minnesota they randomly selected 1238 for training, 306 for validation, and 386 for testing. The model was further tested on external datasets from the Mayo Clinic in Arizona (830 CT scans) and from the Mayo Clinic in Cleveland (396). These datasets included subjects with kidney abnormalities.

They achieved a DSC of 0.94 for cortex segmentation and 0.90 for medulla segmentation on the Minnesota test set. On the two larger test sets they achieved a DSC of 0.93 for cortex and 0.88 for medulla. Furthermore, they found by statistical analysis that differences between manual segmentation and model segmentation were not evident.

4.1.3 Yu et al. (2023)

Yu et al. (2023) [30] proposed a new model called UNesT for 3D medical image segmentation. Different instances of the this model were trained on different datasets, one of which was a dataset consisting of CT scans of the kidneys and corresponding annotations of cortex, medulla and pelvicalyceal system. The design of UNesT can be viewed as a hybrid between a 3D vision transformer and a 3D U-Net. Figure A.1 in appendix A.1 shows the overall architecture. The encoder

is a hierarchical transformer, consisting of nested transformer encoders, and the decoder is similar to the 3D U-Net decoder. First, the input volume is divided into 3D patches of fixed size, which are flattened and projected, producing a 3D grid of feature vectors. Flattening this grid yields a sequence, which is partitioned into non-overlapping blocks of equal length. Each block is then fed through the transformer encoder, and after block-wise processing, the block outputs are de-blockified, i.e. rearranged back into the original 3D grid at that hierarchy level. At the boundary between hierarchies, the spatial resolution is decreased by pooling, which decreases the number of blocks in the next hierarchy. Furthermore, the channel dimension is increased. That is, deeper hierarchies operate on lower resolution but higher-dimensional feature maps, as in the U-Net encoder. Skip connections and a decoder similar to the one in U-Net is used.

The dataset used to obtain cortex and medulla segmentation with UNesT consisted of 116 CT scans with no codes related to kidney dysfunction. They don't explicitly mention that the scans are contrast enhanced, but the data description in their precursor work [34] indicates this. Out of the 116 scans, 96 were used for 5-fold cross validation, and 20 were used for testing. Manual segmentations were verified and corrected independently by expert observers, and for the test set another interpreter performed a second round of manual segmentation. Yu et al. achieved a DSC of 0.92 for cortex segmentation and 0.84 for medulla segmentation.

4.1.4 Chen et al. (2025)

In the study by Chen et al. (2025) [31] three models for cortex and medulla segmentation was implemented, trained, validated and tested: 3D U-Net, originally proposed by Çiçek et al. (2016) [22], Res U-Net, proposed by Zhang et al. (2018) [35], and Swin UNETR, proposed by Hatamizadeh et al. (2022) [36]. They used the nnU-Net framework for choosing training parameters and optimal image preprocessing.

3D Res U-Net is similar to 3D U-Net. The difference is that each convolutional block is replaced by a residual unit. The output of the l 'th residual unit can be described by $\mathbf{x}_{l+1} = f(\mathbf{y}_l)$ with $\mathbf{y}_l = F(\mathbf{x}_l) + \mathbf{x}_l$, where $\mathbf{x}_l$ is the input to the l 'th residual unit, f is an activation function, and F is the residual function, i.e. $F(\mathbf{x}_l)$ is the result of applying the layers that would be in a traditional convolutional block. [35] This modification allows the network to learn residual functions instead of direct mappings. Chen et al. extended the baseline Res U-Net by incorporating atrous convolutions, introduced by Yu and Koltun (2016) [37], and pyramid scene parsing pooling, introduced by Zhao et al. (2017) [38].

Swin UNETR is very similar to UNesT. It has the same structure with a hierarchal transformer as encoder and the usual 3D U-Net decoder. Instead of using blockify and deblockify followed by a pooling block, Swin UNETR uses 3D windows, which are shifted in each hierarchy, and merging layers, where small patches are

concatenated across channel dimension embedding.

464 abdominal CT scans with contrast enhancement obtained from the First Affiliated Hospital of Guangzhou Medical University between 2014 and 2022. The dataset contained subjects with different medical conditions, including kidney stones, hydronephrosis, renal tumors and cysts. Annotations were made by biomedical engineering students in 3D Slicer and reviewed by an instructor and clinical experts. 414 scans were used for training and 50 for testing. The distributions of the training and test set were similar in terms of kidney abnormalities.

With a DSC of 0.9339 for cortex segmentation and 0.8926 for medulla segmentation, Chen et al. achieved a slightly better result with Res U-Net, compared to 3D U-Net, with which they achieved a DSC of 0.9323 for cortex and 0.8911 for medulla. They achieved a DSC of 0.9211 and 0.8776 for cortex and medulla, respectively, using Swin UNETR.

CHAPTER 5

Dataset Description

This chapter provides a description of the two datasets used in this project and how manual segmentation masks were created.

5.1 Data provided by TotalSegmentator

The authors behind TotalSegmentator [29] have made their training dataset of CT scans publicly available. This is a heterogeneous dataset, randomly sampled from 2012, 2016, and 2020 at the University Hospital Basel. It includes subjects with varying ages, genders, and pathologies, as well as scans acquired using different CT scanners, protocols, phases, and sites. For each scan, segmentation masks for the left and right kidneys are provided. These masks were generated by the TotalSegmentator multi-organ model.

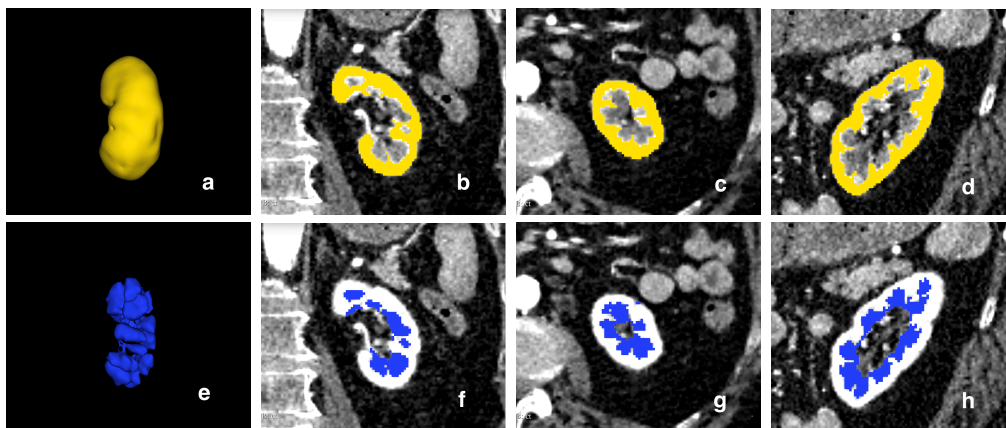


Figure 5.1: Manual annotations of cortex (yellow) and medulla (blue) on a left kidney in a contrast enhanced CT scan, using 3D Slicer with (a, e) in 3D view, (b, f) in coronal view, (c, g) in axial view, and (d, h) in sagittal view. (*Inspired by Xiangling Wang et al.: Age, kidney function, and risk factors associate differently with cortical and medullary volumes of the kidney [39]*).

A subset of the data, consisting of 10 scans, was selected. Among these 10 sub-

jects, ages range from 23 to 85 years; 60% are male and 40% are female, and the average age is 65.7 years. The scans were obtained at four different sites using different protocols.

For all 10 scans, a cortex–medulla segmentation mask of the left kidney was created manually in 3D Slicer, with guidance from a final-year medical student. To construct these masks, voxels within the left whole-kidney mask from TotalSegmentator were defined as the region of interest (ROI), and each voxel was labeled as either cortex or medulla. All other voxels were labeled as background. An example is shown in Figure 5.1, where the annotations in the area around the left kidney are visualized for the subject with ID *s0019*.

5.2 Data provided by Rigshospitalet

The Copenhagen General Population Study (danish: Herlev/Østerbroundersøgelsen) is a large population-based study, established in 2003, that expands upon the Copenhagen City Heart Study (danish: Østerbroundersøgelsen). The Copenhagen City Heart Study began in 1975 and followed residents from the Østerbro district of Copenhagen to investigate coronary heart disease and stroke and related risk factors. Over time, the study expanded to include many other diseases and risk factors. The Copenhagen General Population Study mostly uses the same methods, but expands the scope and greatly increases the number of participants. [40][41][42]

A subset of the data from the Copenhagen General Population Study provided by Rigshospitalet has been used in this project. This subset has 181 CT “full-body” scans and corresponding segmentation masks of the left and right kidney, respectively, generated by the TotalSegmentator multi-organ model. The CT protocol used for this subset is as follows:

1. Non-contrast heart calcium score series
2. Non-contrast thorax for lung series
3. Intravenous contrast injection
4. EKG-gated coronary CTA
5. Abdominal CTA

The abdominal CTA is performed with the remaining contrast after the EKG-gated coronary CTA, and the protocol doesn’t target a specific contrast phase in the abdominal CTA. This means the phase can vary from late arterial to venous, depending on the heart rate of the patient.

For each patient eGFR, blood pressure and creatine level at the time of imaging are available. The dataset is divided into three groups:

1. Patients with low eGFR,
2. Patients with hypertension,
3. Healthy people.

Out of the 181 subjects, 30.5% is in group 1, 29.5% is in group 2, and 40% is in group 3. The eGFR of subjects in group 1 falls within the interval $[37.89, 49.79]$, while the eGFR of subjects in group 3 falls within the interval $[104.6, 135.7]$.

Predicted cortex-medulla-cyst segmentations for subjects in group 1 and 3 will be analysed and compared in chapter 6.

CHAPTER 6

Experiments and Results

This chapter presents the experiments conducted in this project along with their corresponding results. Section 6.1 presents and evaluates a framework for segmentation of cortex, medulla and kidney cysts, while section 6.2 investigates a model for contrast phase classification. Both experiments make use of the two datasets introduced in chapter 5. Section 6.3 focuses on extracting and analysing quantitative imaging biomarkers from the predicted segmentation masks. Finally, section 6.4 describes an additional experiment involving the training of an nnU-Net model.

6.1 Cortex–Medulla–Cyst Segmentation

This section describes and evaluates a framework for cortex, medulla and kidney cyst segmentation, using both datasets described in chapter 5.

6.1.1 Models

The framework uses the Totalsegmentator subtask `kidney_cysts`, which is a single model trained to predict left and right kidney cysts using nnU-Net and a smaller dataset targeted for kidney cyst segmentation. This model has a strongly improved accuracy compared to kidney cyst segmentation with the large multi-organ model.

Furthermore, the framework uses the modified 3D U-Net model introduced by Korfiatis et al. (2022) [5] to obtain segmentations of cortex and medulla. This model predicts five classes: left cortex, right cortex, left medulla, right medulla, and background, and is explained in detail in section 4.1.2. The architecture can be seen in figure 6.1. As explained in section 4.1.2, the model was trained and validated on 1238 and 306 contrast enhanced pre-donation CT scans, respectively. The subjects were healthy individuals evaluated as potential kidney donors, and the scans were acquired prior to donation. Table 6.1 outlines the distribution of the training and validation data.

	Training	Validation
Age	44.5 ± 11.9	43.6 ± 12.1
BMI	28.2 ± 5.0	28.0 ± 5.0
Men	514 (41%)	128 (41%)
Women	724 (59%)	179 (59%)

Table 6.1: Distribution of training and validation data used by Korfiatis et al. (2022) [5]. Age and BMI are expressed as mean $\pm$ standard deviation, and gender distribution is expressed as number and percentage of men and women.

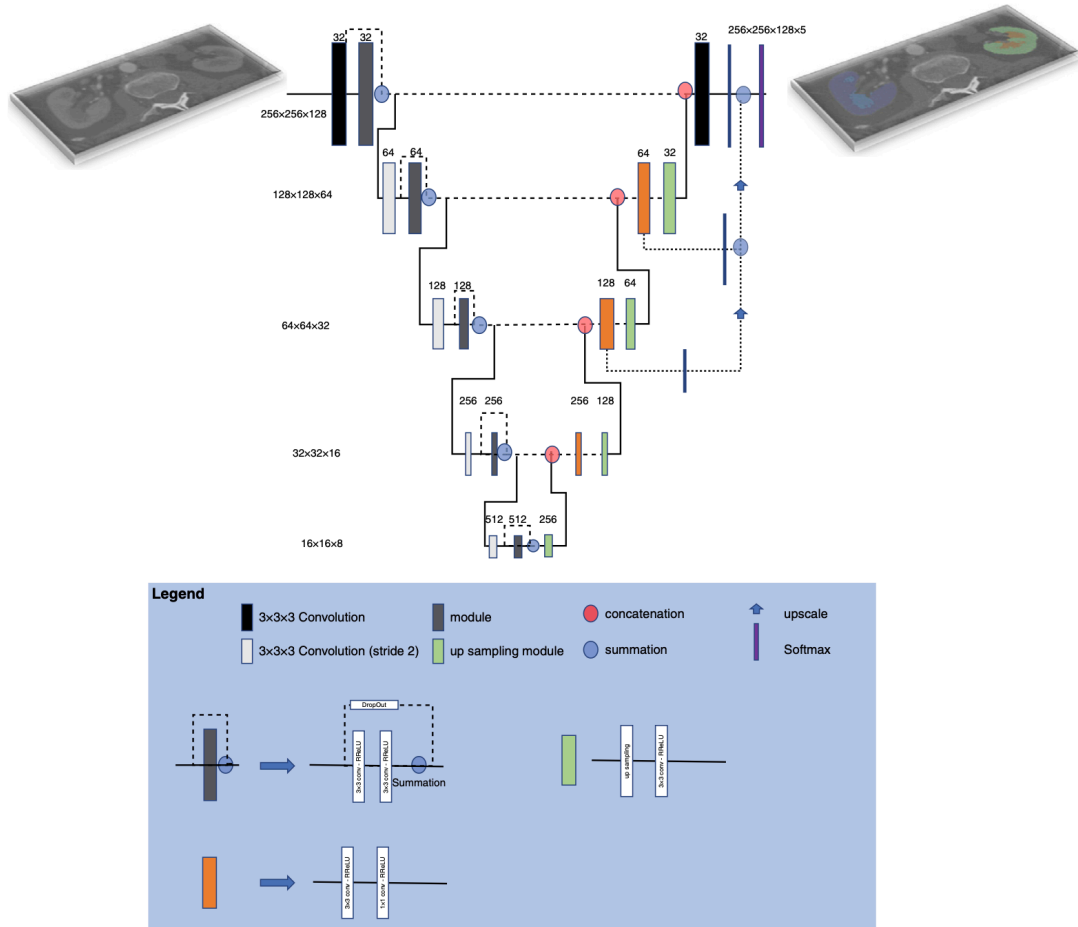


Figure 6.1: Modified 3D U-Net architecture used by Korfiatis et al. for cortex–medulla segmentation in CT scans (*Original from Korfiatis et al. (2022): "Automated Segmentation of Kidney Cortex and Medulla in CT Images: A Multisite Evaluation Study" [5]*).

6.1.2 Framework

This section provides an overview of the processing pipeline, from an initial CT full-body scan to final the generated segmentation mask.

1. Preprocessing

The full-body scan is cropped to the kidney region. To determine the crop, the left and right whole-kidney masks are first merged, and a 3D bounding box is computed as the smallest region enclosing all kidney voxels. This bounding box is then expanded by a 15-voxel margin in every direction to produce the final cropped scan.

2. Raw predictions

Raw kidney-cyst segmentation masks are predicted using the TotalSegmentator tool available on GitHub [43], which performs its own internal preprocessing. Thus, the uncropped full-body scans can be processed directly using the tool, yielding separate left and right cyst masks.

Raw cortex–medulla segmentation masks are obtained using the model and code from the repository by Korfiatis et al. available on GitHub [44]. The cropped scan serves as input, and the script automatically applies additional preprocessing steps such as windowing, normalization, and resizing.

3. Postprocessing

The left and right kidney-cyst masks are merged and cropped to match the spatial extent of the cropped scan. The same crop is applied to the combined whole-kidney mask.

The raw cortex–medulla mask is then restricted using the cropped whole-kidney mask, ensuring that all voxels outside the kidneys are assigned to the background. The left and right cortex classes are merged, as are the left and right medulla classes, yielding a three-class prediction: background, cortex, and medulla.

Finally, this three-class mask is combined with the cropped cyst mask. In cases where a voxel is labelled as both cyst and cortex/medulla, the cyst label takes precedence. The resulting final segmentation mask contains four mutually exclusive classes: background, cortex, medulla, and cyst.

6.1.3 Results

The framework was tested on the two sub-datasets described in chapter 5: a 10-sample dataset from TotalSegmentator, with manual cortex–medulla segmentations, and a 181-sample dataset from the Copenhagen General Population Study.

10-sample dataset

The 10-sample dataset has manual segmentations (acting as ground truth) of cortex and medulla in the left kidney. No cysts were present in the left kidneys, so this class is not present here. When evaluating the framework on the dataset only predictions for cortex, medulla and cysts in the left kidney are considered. As an evaluation metric the Dice score (DSC) per class is used. The DSC results can

be seen in table 6.2, while an example of prediction and ground truth on a single subject can be seen in figure 6.2.

	Cortex	Medulla
s0019	0.9044	0.8011
s0065	0.9114	0.8623
s0371	0.9036	0.8354
s0392	0.8723	0.8049
s0403	0.9163	0.7879
s1285	0.9009	0.8440
s1314	0.8479	0.5396
s1349	0.9138	0.7811
s1372	0.8502	0.8012
s1382	0.8886	0.8290

Table 6.2: DSC for cortex and medulla across subjects in the 10-sample dataset.

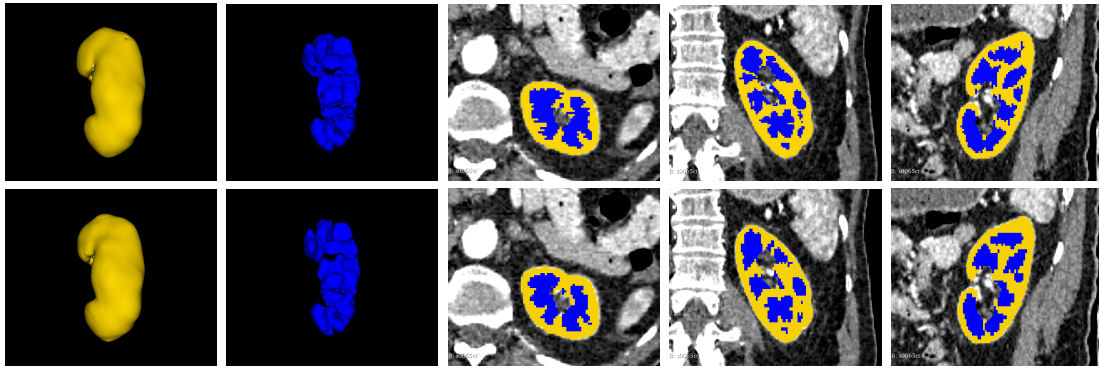


Figure 6.2: Ground truth and predicted segmentation for subject s0065 in the 10-sample dataset. In the first row ground truth for cortex (yellow) and medulla (blue) is shown in 3D view, axial view, coronal view, and sagittal view, respectively. In the second row the predictions for cortex (yellow) and medulla (blue) are shown in corresponding views.

As seen in Table 6.2, the DSC is generally higher for cortex than for medulla. This aligns with results from the literature described in chapter 4. Furthermore, there is a clear difference in performance for the different subjects. Excluding the obvious outlier (s1314), the DSC is varying from 0.8502 (s1372) to 0.9163 (s0403), with an average of 0.8957, for cortex and from 0.7811 (s1349) to 0.8623 (s0065), with an average of 0.8163, for medulla. Subject s1314 has a particularly low DSC of 0.5396 for medulla and also the lowest DSC of 0.8479 for cortex.

181-sample dataset

Since there are no manual segmentations available for cortex, medulla and kidney cysts in the 181-sample dataset, a quantitative evaluation of the framework cannot be made on this dataset. Instead, the predictions are inspected visually. Examples

can be seen in figure 6.3 and in figure 6.4.

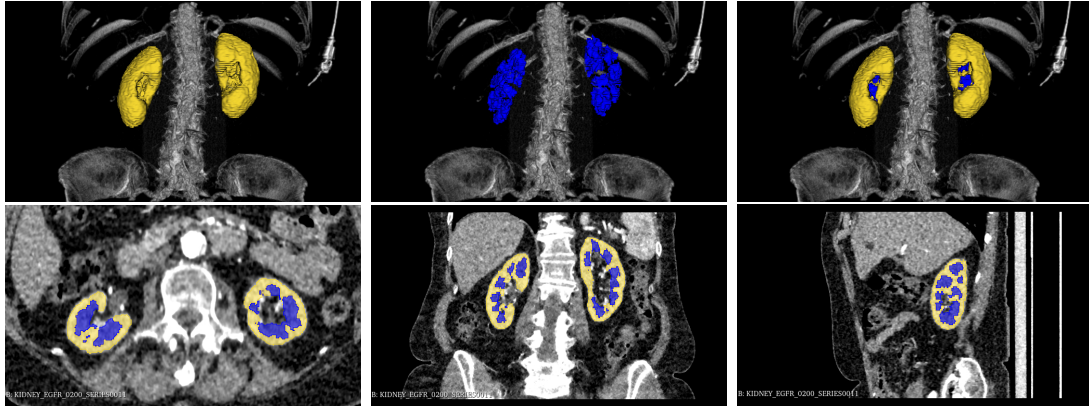


Figure 6.3: Cortex (yellow) and medulla (blue) prediction on a scan from the 181-sample dataset showing a patient with low eGFR. In the top row the predictions are seen in 3D view, while the predictions are seen in axial, coronal, and sagittal view (from left to right) in the bottom row.

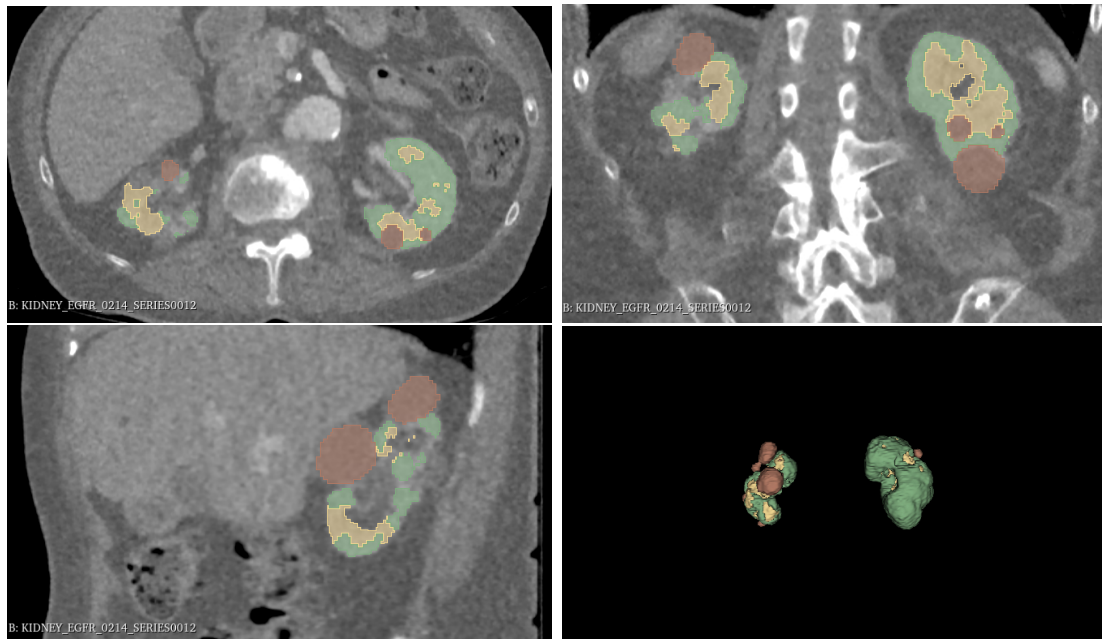


Figure 6.4: Cortex (green), medulla (yellow) and cyst (brown) prediction on a scan from the 181-sample dataset showing a patient with low eGFR. In the top row the predictions are seen in axial and coronal view, and in the bottom row the predictions are seen in sagittal and 3D view.

The predicted segmentation masks appear realistic. The scan in figure 6.4 contains multiple cysts, whereas the other scan in figure 6.3 shows none. In figure 6.3 cortex appropriately surrounds medulla, and medulla exhibits the expected pyramid-like structures. However, in figure 6.4, it can be seen that medulla is not fully enclosed by cortex, as it usually is. Furthermore, some relatively large

parts of the kidney, possibly cortex, have likely been misclassified as background, especially in the right kidney (displayed on the left side of the image). Most the scans in the dataset yield segmentations similar to the one in figure 6.3, but some have larger, potentially misclassified volumes, like the case in figure 6.4. By inspection, these issues appear to occur more frequently when cysts are present, and often in the right kidney.

In total, cyst(s) in the left and/or right kidney were detected on 38 out of the 181 scans. Out of these 17 scans were of patients with low eGFR, 4 scans were of healthy patients, and the remaining 17 scans were of patients with hypertension.

6.2 Contrast Phase Detection

The method of predicting the contrast phase present in a CT scan, and the results on the two sub-datasets described in chapter 5, are presented in this section.

6.2.1 Model

A TotalSegmentator tool available on GitHub [43] is used to predict contrast phases from full-body CT scans. The model behind the tool is an ensemble of five XGBoost regressors, which was trained on the large TotalSegmentator dataset as follows.

First, they used the TotalSegmentator multi-organ model to predict 20 different anatomical structures in all of the scans in the dataset. Most of these structures are found in the abdominal area, but also other organs such as the heart and brain are included. Based on the predictions the median HU value of each structure within each scan was computed. These values were then used as features for training an XGBoost regressor to predict the post injection time in a CT scan.

XGBoost stands for (Extreme Gradient Boosting) and is an implementation of gradient tree boosting, which is explained in section 3.2. The XGBoost regressor was trained using five-fold cross-validation. Therefore, in reality five different models were obtained, and the final prediction of the post injection time when running inference (`pi_time`) is then the mean predicted post injection time. The standard deviation (`stddev`) of the five predictions is reported as well and can be used as a measure of confidence. Furthermore, the minimum and maximum post injection time predicted (`pi_time_min` and `pi_time_max`) are reported.

The predicted post injection time is then used to finally classify a CT scan into one of four contrast phases: native (pre-contrast), early arterial, late arterial, and portal venous. When a scan is classified the output contains a probability (`prob.`) for the predicted class, which is high if the predicted post injection time is close to the ideal post injection time for the given phase.

The results they obtained from five-fold cross validation were a mean absolute error of 5.55 seconds on the post injection time, and F1 scores of 0.980 (native), 0.915 (early and late arterial), and 0.940 (portal venous) for the four classes.

6.2.2 Results

The model was used to predict contrast phases in both datasets described in chapter 5. The exact phases in both datasets are unknown, although the phases in the 181-sample dataset are expected to range from late arterial to portal venous. The results are outlined in the following.

10-sample dataset

	pi_time	phase	prob.	pi_time_min	pi_time_max	stddev
s0019	28.74	arterial_early	1.0	26.17	30.62	1.8680
s0065	34.52	arterial_late	1.0	29.40	37.39	3.2937
s0371	38.82	arterial_late	1.0	35.47	40.93	1.9624
s0392	28.81	arterial_early	1.0	27.51	31.85	1.6275
s0403	53.95	arterial_late	0.7	51.67	57.26	1.8221
s1285	26.19	arterial_early	1.0	24.08	27.46	1.1344
s1314	66.15	portal_venous	1.0	65.15	67.08	0.8138
s1349	28.59	arterial_early	1.0	26.65	31.29	1.9458
s1372	41.83	arterial_late	1.0	38.47	46.89	2.7761
s1382	44.00	arterial_late	1.0	39.53	48.00	2.9300

Table 6.3: Results from the TotalSegmentator phase prediction model on the 10-sample dataset. Times are measured in seconds.

The 10-sample results in Table 6.3 show that the predicted contrast phases for all subjects in the dataset—except for subject s1314—are arterial phases. Of these arterial predictions, five correspond to late arterial and four to early arterial phases. The predicted phase for subject s1314 is portal venous.

Furthermore, the model assigns probability 1 to all phase predictions except for the late arterial prediction for subject s0403, which is assigned probability 0.7. This corresponds with the very high post injection time of approximately 54 seconds compared to the other predicted late arterial post injection times.

Predicted early arterial post injection times cluster tightly around 26–29 seconds, while late arterial timings span a wider range of 34–44 seconds, excluding subject s0403, which is likely an outlier. The standard deviation is generally low for early arterial cases, and especially for the portal venous case, but higher for several late arterial predictions.

181-sample dataset

	native	arterial_early	arterial_late	portal_venous
Count	0	1	164	16

Table 6.4: Number of phase predictions in each class for the 181-sample dataset.

As seen in 6.4 most of the scans are predicted to display a late arterial contrast phase, while a few are predicted to display a portal venous phase. Early arterial phase is only predicted for a single scan, and no native phase is predicted. This aligns with the CT protocol described in section 5.2, where the phases are expected to range from late arterial to portal venous.

The probability outputs for the phases on the dataset were always either 1 or 0.7. For the single early arterial phase and for all portal venous phases the assigned probability was 1. Among the late arterial phases, 62 phases we assigned probability 0.7, while 119 phases were assigned probability 1.

The predicted post-injection times vary more for subjects, where the late arterial phase was predicted, compared to those where the portal venous phase was predicted. In addition to this, the standard deviation across the five-model ensemble predictions was generally higher for subjects with a late arterial phase prediction—consistent with what was observed in the 10-sample dataset. Boxplots can be seen in figures A.2 and A.3 in the appendix.

6.3 Volumetric Biomarker Analysis

This section presents and analyses relevant volumes extracted from the segmentations obtained from the 181-sample dataset in section 5.2, using the framework described in section 6.1. All statistical tests presented in the following are performed using the `scipy.stats` implementations. After using the voxel spacings to find the unit-voxel volume, the following volumes and ratios are computed from the segmentations:

- Total cortex volume in mm^3 (TCOV)
- Total medulla volume in mm^3 (TMV)
- Total cyst volume in mm^3 (TCYV)
- Total cortex–medulla volume ratio (TCMVR)
- Total cyst volume fraction (TCVF)
- Asymmetry (A)

Total refers to the fact that the volumes in both kidneys are computed. Total cortex–medulla ratio is defined as

$$\text{TCMVR} := \frac{\text{Total cortex volume}}{\text{Total medulla volume}}.$$

Total cyst volume fraction is defined as

$$\text{TCVR} := \frac{\text{Total cyst volume}}{\text{Total cortex volume} + \text{total medulla volume} + \text{total cyst volume}}.$$

Asymmetry, which the whole kidney masks have been used to compute, is in this report defined as

$$A := \frac{\text{Volume of smallest kidney}}{\text{Volume of largest kidney}}.$$

The following analysis examines whether the considered volumes and ratios differ significantly between a group of patients with low eGFR and a group of healthy controls. These groups are described in section 5.2. The distributions of the variables in each group are visualized in figures 6.5 and 6.6.

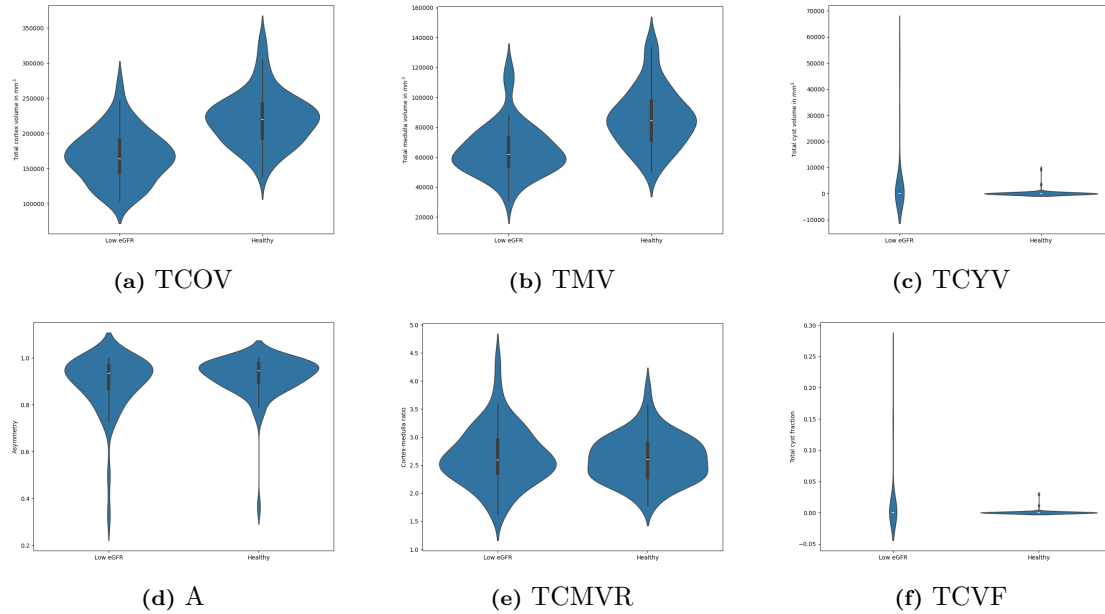


Figure 6.5: Violin plots of variables in the 181-sample dataset within the group of patients with low eGFR (left) and the healthy control group (right).

Note that the violin plots in figures 6.5c, 6.5f, and 6.5d, suggest that there are samples with negative values of total cyst volume (TCYV) and total cyst volume fraction (TCVF), and samples with asymmetry measure above 1. Naturally, this is not possible. However, the violin plots only show the approximated distribution, not the real data. Compared to the box plots in figures 6.6c, 6.6f, and 6.6d, we see that the real data for TCYV and TCVF are in fact non-negative, and the real data for the asymmetry are upper bounded by 1.

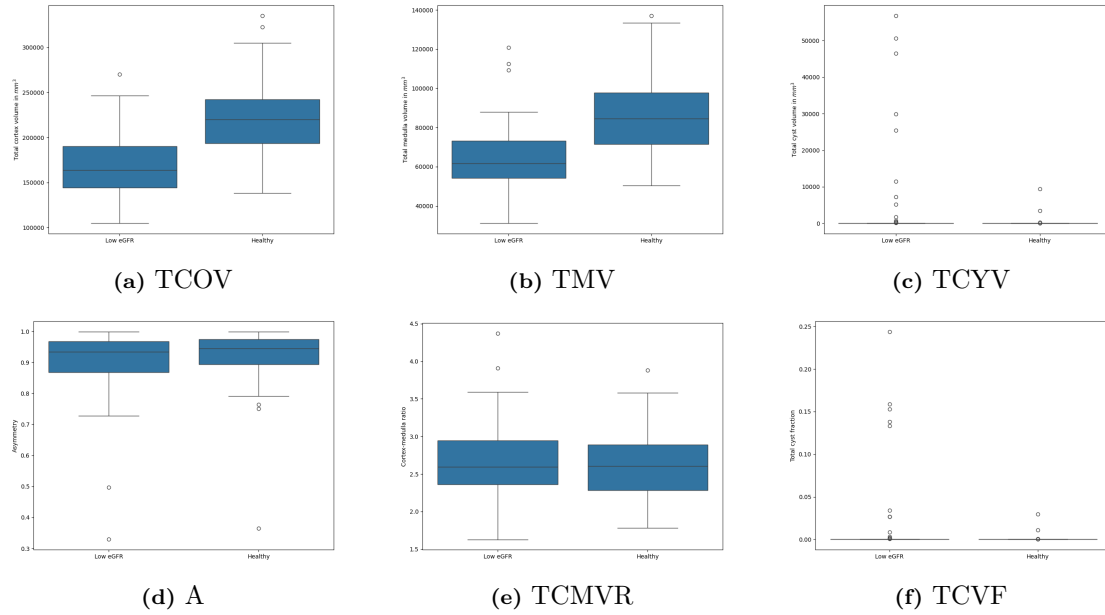


Figure 6.6: Box plots of variables in the 181-sample dataset within the group of patients with low eGFR (left) and the healthy control group (right).

Figures 6.5c and 6.5f show that most subjects in both groups have no cysts (volumes and fractions equal to zero). However, several subjects—particularly in the low eGFR group—exhibit larger cyst volumes and fractions.

Normality Testing

As a first step, the distributions within each group were assessed using the Shapiro–Wilk test, described in section 3.4, with the resulting p -values shown in table 6.5. The test indicated significant deviation from normality at the 5% level – and in most cases even at the 0.1% level – for all variables except total cortex volume and total cortex–medulla volume ratio, in both groups, as well as total medulla volume in healthy controls. Therefore, these exceptions will be regarded as normally distributed.

	Low eGFR, p -value	Healthy, p -value
TCOV	0.436	0.114
TMV	$1.55 \times 10^{-3**}$	0.0578
TCYV	$0.889 \cdot 10^{-13***}$	$0.206 \cdot 10^{-17***}$
TCMVR	0.103	0.406
TCVF	$1.01 \cdot 10^{-13***}$	$2.08 \cdot 10^{-18***}$
A	$1.14 \cdot 10^{-8***}$	$2.76 \cdot 10^{-11***}$

Table 6.5: p -values from the Shapiro–Wilk test for the two groups (patients with low eGFR and healthy controls). Two stars indicate a p -value below 0.01, and three stars indicate a p -value below 0.001.

Group Differences

Next, in order to examine whether a significant difference between the distribution of variables in the two groups can be established, Welch’s t -test and Mann–Whitney U -test were performed. The statistical tests are explained in section 3.4. If the data for a variable could be considered normally distributed, based on the result from the Shapiro–Wilk test, within both groups, Welch’s t -test was performed. This specific t -test was chosen since the variances are not equal between groups. Otherwise, the Mann–Whitney U -test was performed. The resulting p -values from the tests can be seen in table 6.6.

	p -value from Welch’s t -test	p -value from Mann–Whitney U -test
TCOV	$1.17 \times 10^{-12***}$	–
TMV	–	$1.03 \times 10^{-9***}$
TCYV	–	$1.37 \cdot 10^{-4***}$
TCMVR	0.599	–
TCVF	–	$1.31 \cdot 10^{-4***}$
A	–	0.250

Table 6.6: p -values from the t -test and Mann–Whitney U -test for the two groups (patients with low eGFR and healthy controls). Three stars indicate a p -value below 0.001.

Based on the p -values reported in table 6.6, no statistically significant differences were found between the two groups for the total cortex–medulla volume ratio or for asymmetry. For both variables, the null hypotheses of Welch’s t -test (equal means) and the Mann–Whitney U -test (equal distributions) could not be rejected. However, the null hypotheses were rejected on a significance level of 0.1% for all other variables, indicating a statistically significant difference between the distributions of the two groups for these variables.

The p -values for total cortex volume and total medulla volume are particularly low, indicating the strongest statistical evidence for between-group differences in these two variables. This corresponds with what we would expect based on figures 6.5 and 6.6, where a clear shift in the distributions of total cortex volume and total medulla volumes is seen between the groups. Given the larger number of subjects with extremely high cyst volume and fraction in one of the groups, which can be seen in figures 6.6c and 6.6f, differences in the distributions of these variables between the two groups are expected as well.

6.4 Segmentation Using nnU-Net

6.4.1 Model

As a final experiment, an additional 3D U-Net model for cortex–medulla–cyst segmentation was trained and validated using the latest version (v2) of the nnU-Net framework [45]. The model was trained with five-fold cross-validation on the

181-sample dataset described in section 5.2. Because no manual ground truth segmentations are available, the predicted masks generated by the framework described in section 6.1 were used as training labels. By design, all samples were included in the cross-validation setup, and no independent test set was held out. The nnU-Net framework automatically determines the configuration for different model types based on the nature of the dataset. In this case the configuration corresponding to the 3D residual encoder U-Net, using full resolution, was chosen.

The model uses seven encoder-decoder stages, progressively increasing feature channels from 32 to 320. Convolutions are 3D, mostly $3 \times 3 \times 3$ kernels, with instance normalization and leaky ReLU activations. Strides vary per stage, allowing downsampling and the decoder has fewer convolutions per stage. A patch size of [80, 256, 160] and batch size of 2 was used. After training concludes, full-volume predictions on the validation fold are generated via sliding-window inference, and Dice scores are computed per class and per sample. Learning rate, optimizer and loss function are the standard, fixed nnU-Net parameters. Thus, the learning rate decays linearly from 0.01 to 0 throughout epochs, and the loss function is a combination of soft Dice loss and CE.

During preprocessing the CT scans were normalized, and the standard nnU-Net data augmentation was applied. However, mirroring was explicitly disabled. Right and left kidney anatomy differ naturally, since for instance the right kidney is placed lower, and the organs surrounding each kidney are different. Therefore, mirroring would create unrealistic samples and could confuse the model by placing right-kidney anatomy on the left side and vice versa, leading to worse performance.

6.4.2 Results

Table 6.7 shows the results of the five-fold cross-validation. The model performance for cortex and medulla segmentation is consistent across folds. The performance for cortex segmentation is generally better with an average DSC of 0.934 on all folds compared to 0.898 for medulla segmentation. The model performs poorly for cyst segmentation with DSC ranging from 0.056 to 0.323 across folds and an average DSC of 0.201 on all folds.

	Cortex	Medulla	Cyst	Foreground
Fold 0	0.935	0.898	0.220	0.684
Fold 1	0.933	0.897	0.211	0.681
Fold 2	0.936	0.901	0.056	0.631
Fold 3	0.935	0.901	0.193	0.676
Fold 4	0.932	0.895	0.323	0.717
All folds	0.934	0.898	0.201	0.678

Table 6.7: nnU-Net validation results (mean DSC) from five-fold cross-validation. Foreground classes consist of cortex, medulla and cyst. The scores for all folds are the averages of the class scores across all folds.

CHAPTER 7

Discussion

This chapter discusses the main findings of the project and evaluates how the chosen methods influenced the results. The discussion focuses on the impact of data variability, limitations in the modelling approach, challenges related to defining ground truth in medical imaging, and the origins of dependence underlying the statistically significant differences observed in the analysis. Relevant future work is discussed as well.

The Influence of Contrast Phase in Cortex–Medulla Segmentation

The Dice scores on the 10-sample dataset, obtained using the framework described in Section 6.1, were generally high and consistently higher for cortex segmentation than for medulla segmentation. Excluding subject s1314, the DSCs ranged from 0.8502 to 0.9163 for cortex segmentation and from 0.7811 to 0.8623 for medulla segmentation. For subject s1314, however, the DSCs dropped to 0.8479 for cortex segmentation and 0.5396 for medulla segmentation. Phase prediction for the 10 scans indicated that all but one were classified as arterial (early or late). Only the scan of subject s1314 was predicted to be in the portal venous phase. This suggests that the segmentation model employed in the framework struggles with scans in this phase—an expected outcome, as the cortex–medulla boundary is less distinct in non-arterial phases.

The predictions on the 181-sample dataset generated by the framework were generally realistic. However, some segmentations showed that the medulla was not fully enclosed by the cortex, and larger regions appeared to be misclassified as background—particularly in cases with cysts and more frequently in the right kidney. Several factors may account for these observations. First, a substantial proportion of the scans in the 181-sample dataset were predicted to be in the portal venous phase. As observed in the 10-sample dataset, scans in this phase are probably more challenging for the model because the intensity difference between cortex and medulla is reduced, making the boundary less distinct. Second, the datasets include subjects with very low eGFR, where the cortex is likely to be

markedly thinned. Thin cortical tissue may be difficult to visualise at lower resolutions, and downsampling in the network could make these structures even harder to preserve, which may explain why the medulla sometimes appears unenclosed. Third, inaccuracies in the whole-kidney masks can affect post-processing: if the foreground restriction is imperfect, cortex voxels located near the kidney boundary may be incorrectly removed. In addition, the example scan published together with the model by Korfiatis et al. [5] was not stored in the correct orientation. If the model was trained on similarly disoriented images, this could introduce left-right mismatches that result in segmentation errors when applying the model to predict segmentations for correctly oriented scans. Strategies for improving segmentations in cases like those described above are discussed in the "Future Work" section below.

Finally, and perhaps most importantly, the segmentation errors may be related to the substantial differences between the nature of the training data used for the model (Korfiatis et al.) applied in the framework and the datasets investigated in this project.

Difference Between Training and Test Data

Firstly, the model developed by Korfiatis et al. was trained exclusively on contrast-enhanced arterial-phase CT scans, whereas several cases in the datasets used in this project are predicted to be in non-arterial phases. The demographic characteristics also differ substantially. The training data consisted of comparatively young subjects with a narrow age distribution (44.5 ± 11.9 years), while the 10-sample dataset spans a much wider range (23–85 years) with a considerably higher mean age (65.7 years). The subjects in the 181-sample dataset should also generally be older, although a detailed age distribution is not provided in this report. Moreover, the training data contained only healthy individuals without kidney abnormalities. In fact, the subjects were potential kidney donors and can therefore be expected to have above-average kidney function and high eGFR values. This stands in clear contrast to the 181-sample dataset, in which 60% of subjects were not healthy and 30.5% exhibited very low eGFR.

Because cortical thickness decreases with both increasing age and declining kidney function, the kidneys—and potentially the image characteristics—are likely to differ notably between our datasets and the data used to train the model. Similarly, the contrast can be expected to differ due to different phases. Such differences may constitute a domain shift that the model cannot fully compensate for, which could explain the instances of incomplete or missing segmentations observed.

It should be noted that Korfiatis et al. reported high DSCs of 0.93 and 0.88 for cortex and medulla segmentation, respectively, on external test sets that included cases with kidney abnormalities. However, the nature of these abnormalities was not specified, and it is possible that the external scans were still similar to the

training data with respect to key factors such as contrast phase.

Origin of Significant Difference in Volumes

By inspection, the implemented segmentation framework appeared to produce incomplete segmentations more frequently when cysts were present. Because cysts occurred substantially more often in the low-eGFR group than in the healthy controls, an important question arises: do the statistically significant differences in, for example, total cortex and medulla volumes reflect true anatomical differences, or are they influenced by systematic under-segmentation in subjects with low eGFR? This possibility is plausible, as the cortex–medulla model was trained on healthy donor candidates who were unlikely to have cysts, which could reduce model performance in cystic kidneys. In addition to cyst-related effects, the exclusive use of healthy kidneys in training may contribute to further under-segmentation in the low-eGFR group due to general anatomical differences such as a thinner cortex. Furthermore, since the specific ages of the subjects in the 181-sample dataset were not available for this project, it is possible that the differences in volumes between the low eGFR group and healthy controls may also reflect age differences and not just differences in eGFR.

Defining Ground Truth in Medical Imaging

Inspection of the manual annotations in the 10-sample dataset, used as ground truth, alongside the corresponding predicted segmentations shows that the predictions are noticeably smoother and, in many cases, appear more anatomically plausible than the manual annotations. It is therefore worth considering whether the DSC reflects the performance accurately in these cases. Furthermore, it raises an important question regarding how ground truth should best be defined. The manual segmentations in this project were not produced by a medical professional, which likely affects their quality. However, even for expert annotators it can be very challenging to segment 3D data, particularly when anatomical boundaries are not always clear, which is often the case with the cortex-medulla boundary. If model predictions can sometimes exceed the quality of the manual annotations, there is a risk that training models on such annotations will replicate human errors rather than refining structures.

Evaluation of Phase Predictions

The phase predictions on the 181-sample dataset were in correspondance with the expected contrast phases based on the CT protocol, which indicate an overall good model performance. The mapping from predicted post-injection time to contrast phase was generally assigned high probability on both datasets—most frequently 1 and otherwise 0.7, which is good. However, the fact that the outputted probabilities were discrete could indicate that the mapping from predicted post-injection time to probability likely isn’t nuanced enough to represent uncertainty accurately,

which should be taken into account.

Early arterial and portal venous predictions had more tightly clustered predicted post-injection times across samples compared to late arterial. In addition, the results highlight differences in prediction stability across phases. The early arterial and portal venous predictions generally had slightly lower ensemble standard deviation and were always assigned probability 1, compared to the late arterial predictions with higher standard deviation and a large number of classifications with probability 0.7. These findings suggest that the late arterial phase is more variable, and that the model had more difficulty producing consistent post-injection time estimates for subjects predicted to be in this phase.

Finally, it should be noted that the scans in this dataset are not really "full"-body scans, but abdominal scans. Therefore, not all features used to train the XGBoost regressor can be computed; for example, the mean HU value of the brain cannot be computed. This limitation could make predicting the correct post-injection time more difficult and may in some cases contribute to inaccuracies in phase classification.

Cortex–Medulla–Cyst Segmentation and nnU-Net

The results of five-fold cross validation show very similar performance across all folds for cortex and medulla segmentation and high average DSCs of 0.934 and 0.898, respectively. In contrast the performance for cyst segmentation varies substantially across folds with low DSCs and an average DSC of 0.201. A likely explanation is that the cyst class was present in only 38 of the 181 labels. Furthermore, many subjects with cysts had only one or a few very small cysts, whereas others had multiple large cysts. This led to highly imbalanced training data, which the model likely struggled to handle.

It should be noted that, although the results appear to demonstrate strong model performance for cortex and medulla segmentation, the training was not based on true ground truth but rather on the segmentation outputs generated by the framework described in Section 6.1. Nevertheless, the observed performance suggests promising potential for training an accurate model using actual ground truth data.

7.1 Future Work

The results of this project open several opportunities for future research. First of all, it would be interesting to train an nnU-Net on a larger dataset with actual ground truth data, since the method showed promising results. This dataset should be balanced regarding both age distribution and abnormalities. Furthermore, it could be considered to add class weights to compensate for the small size of the cyst class, although one should be careful to prevent overfitting.

The specific age distribution in the 181-sample dataset should be investigated, especially within the two groups where volumes were compared. More work could also be put into correcting the segmentation masks produced by the framework in section 6.1 by for instance disregarding background predictions within whole-kidney masks and instead selecting the foreground class with highest probability. In addition, it could be interesting to test whether flipping the images in preprocessing would improve segmentations if the applied model was in fact trained on disoriented images. Finally, it might be interesting in general to divide classes into left and right and analyse the kidneys and sub-volumes separately, as it might not significantly impact the renal system if just a single kidney is damaged.

CHAPTER 8

Conclusion

The segmentation combining the TotalSegmentator tool for cyst segmentation and an existing 3D U-Net from Korfiatis et al. [5] generally achieved high performance for cortex and medulla segmentation on the 10-sample dataset. DSCs around 0.8-0.9 were obtained, comparable to results reported in the literature. Also in line with the literature, cortex segmentation achieved a higher DSC compared to medulla segmentation. A significant performance drop was observed for a scan predicted to be in the portal venous phase. This observation suggests a substantial influence of contrast phase on cortex–medulla segmentation quality.

Applying the framework to the 181-sample dataset produced segmentation masks that visually seemed anatomically realistic. However, in some cases incomplete cortex–medulla separation or regions possibly misclassified as background were observed, especially in scans containing cysts and more frequently in the right kidney. These errors are likely related to differences between the data used to train the cortex-medulla segmentation model and the dataset analysed in this project. These differences include for instance contrast phase, subject age, kidney function, and the presence of structural abnormalities. Such domain shifts seem to limit the robustness of the pretrained model by Korfiatis et al.

Contrast phase predictions obtained from the XGBoost based tool from TotalSegmentator were consistent with the protocols for the 181-sample dataset. Most scans were classified as late arterial, subsequently portal venous, while the 10-sample dataset was in all but one case predicted to be early and late arterial. The predicted post-injection times and ensemble standard deviations indicated that the late arterial phase is more variable and difficult to estimate than early arterial or portal venous phases. Although the phase prediction model generally assigned high confidence to its predictions, the discreteness of probability outputs suggests that uncertainty might not be fully captured.

Using the predicted segmentations, several volumetric biomarkers were extracted and statistically compared between patients with low eGFR and healthy controls.

Statistically significant differences were found on a 0.1% level for total cortex volume, total medulla volume, total cyst volume, and total cyst volume fraction. No significant differences were observed for total cortex–medulla volume ratio or kidney asymmetry. While these findings generally align with the hypothesis, they must be interpreted carefully. The segmentation framework might have under-segmented in the case of cystic kidneys and other pathological image characteristics, and age differences between groups could not be taken into account.

The nnU-Net experiment demonstrated high and stable performance for cortex and medulla segmentation based on cross-validation with mean DSCs of 0.93 and 0.9 for cortex and medulla, respectively. However, cyst segmentation performance was unstable and poor with a mean DSC of 0.2, likely due to class imbalance and large variability in cyst size and frequency. Although these results only hold on the pseudo-ground truth, they do indicate a promising possibility to train a single well-performing model on actual ground truth.

In conclusion, the findings in this project highlight both potential and limitations of AI-based kidney substructure segmentation from CT scans. While existing models can produce reasonable segmentations and enable large-scale analysis, their performance might be strongly influenced by contrast phase, and demographic and pathological differences. Future work using larger and more diverse datasets with high-quality ground truth annotations is needed to improve general performance and obtain a more reliable between-groups analysis of the extracted segmentation volumes.

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Appendix

A.1 UNesT Model Architecture

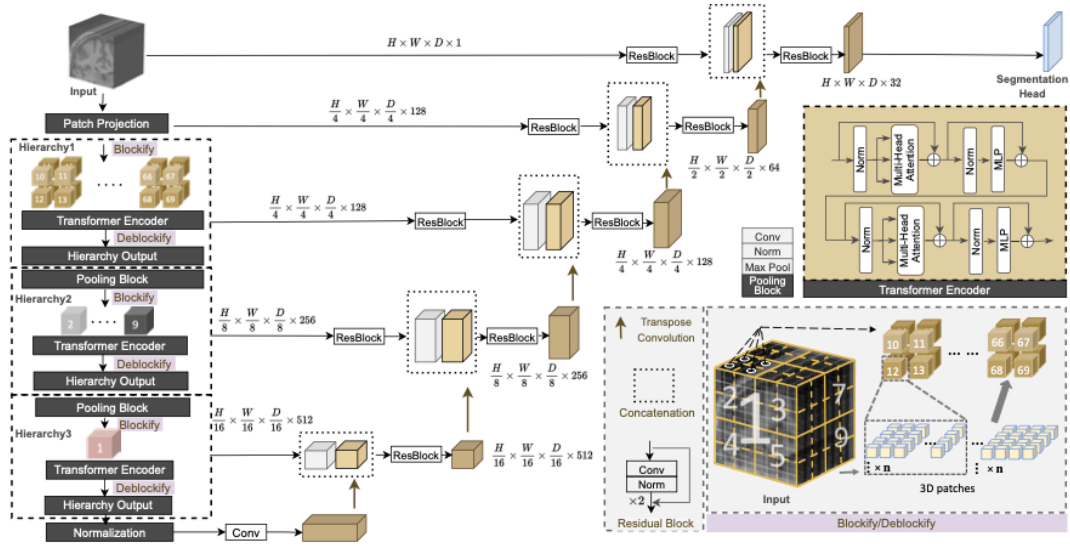


Figure A.1: UNesT architecture used by Yu et al. (2023) for 3D medical image segmentation from CT scans (*Original from Yu et. al (2023): "UNesT: Local Spatial Representation Learning with Hierarchical Transformer for Efficient Medical Segmentation" [30]*).

A.2 Distribution of Post-Injection Times

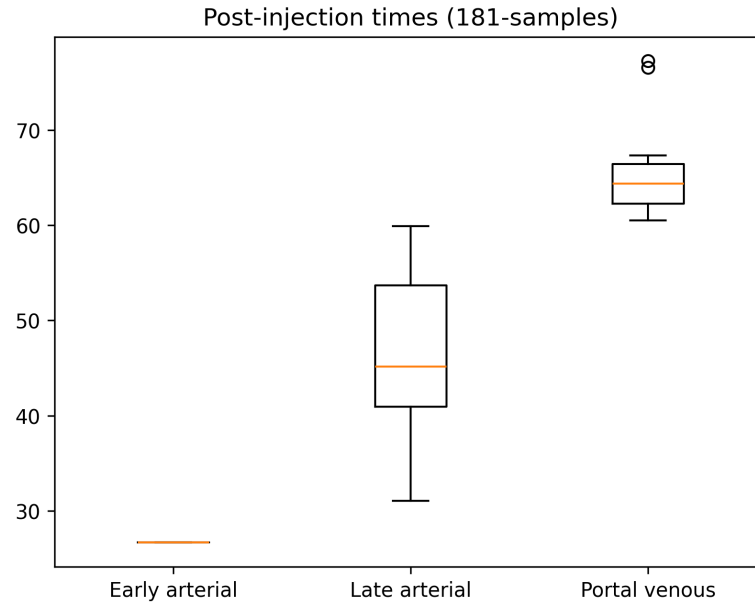


Figure A.2: Final post-injection time predictions across samples in the 181-sample dataset. Samples are grouped according to the predicted contrast phases, and times are measured in seconds.

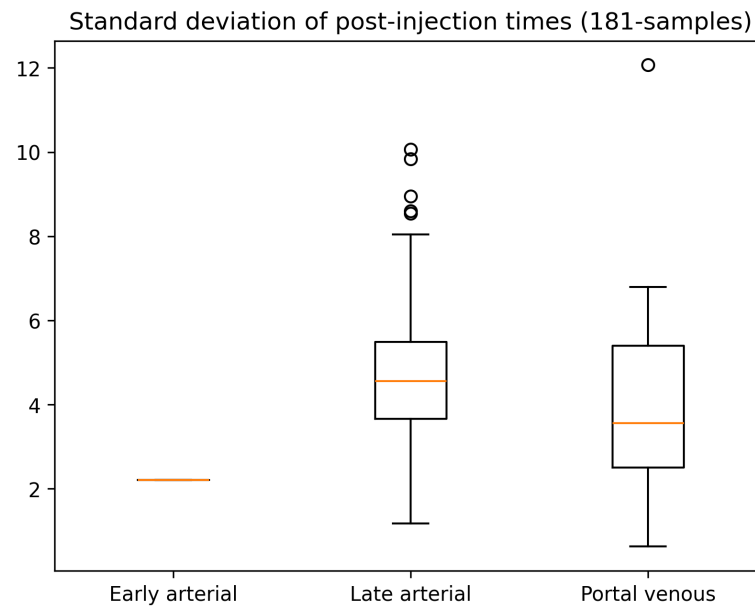


Figure A.3: Standard deviation for the ensemble predictions of post injection time for each sample in the 181-sample dataset. The samples are grouped according to the predicted contrast phases, and standard deviation is measured in seconds.